

Clethodim 2E

HERBICIDE

ACTIVE INGREDIENT:

Clethodim: (E)-2-[1-[(3-chloro-2-propenyl)oxy]imino]

propyl]-5-[2-(ethylthio)propyl]-3-hydroxy-2-cyclohexen-1-one.....26.4%

OTHER INGREDIENTS:.....73.6%**TOTAL**.....100.0%

Contains petroleum distillate. Contains 2.0 lbs. clethodim per gallon.

KEEP OUT OF REACH OF CHILDREN WARNING – AVISO

Si usted no entiende la etiqueta, busque a alguien para que se la explique a usted en detalle.
(If you do not understand the label, find someone to explain it to you in detail.)

FIRST AID**If in eyes:**

- Hold eye open and rinse slowly and gently with water for 15-20 minutes.
- Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye.
- Call a poison control center or doctor for treatment advice.

If on skin or clothing:

- Take off contaminated clothing.
- Rinse skin immediately with plenty of water for 15-20 minutes.
- Call a poison control center or doctor for treatment advice.

If swallowed:

- Immediately call a poison control center or doctor.
- Do not induce vomiting unless told to do so by a poison control center or doctor.
- Do not give any liquid to the person.
- Do not give anything by mouth to an unconscious person.

If inhaled:

- Move person to fresh air.
- If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible.
- Call a poison control center or doctor for further treatment advice.

HOT LINE NUMBER

Have the product container or label with you when calling a poison control center or doctor or going for treatment. You may also contact the Poison Control Center at **1-800-222-1222** for emergency medical treatment information. For non-emergency information concerning this product, contact the National Pesticide Information Center (NPIC) at 1-800-858-7378, Monday through Friday 8AM to 12PM Pacific time, or at <http://npic.orst.edu>.

NOTE TO PHYSICIAN: Ingestion of this product or subsequent vomiting can result in aspiration of light hydrocarbon liquid, which can cause pneumonitis. Contains petroleum distillate.

For Chemical Emergency:

Spill, Leak, Fire, Exposure, or Accident, Call CHEMTREC Day or Night

Within USA : 1-800-424-9300 or +1 703-527-3887 (collect calls accepted)

See label booklet for additional Precautionary Statements, First Aid and Directions for Use including Storage and Disposal instructions.

Manufactured for:

Tenko, Inc.

1725 Windward Concourse, Suite 410

Alpharetta, GA 30005

EPA Reg. No.: 103591-11-55467

EPA Est. No.: 84662-CHN-1

Net Contents: 2.5 GAL

MADE IN CHINA

PRECAUTIONARY STATEMENTS
HAZARDS TO HUMANS AND DOMESTIC ANIMALS
WARNING

Causes substantial but temporary eye irritation. Avoid contact with skin. Do not get in eyes, on skin, or on clothing. Harmful if swallowed or inhaled. Avoid breathing vapors or spray mist. Prolonged or frequently repeated skin contact may cause allergic reactions in some individuals. Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco, or using the toilet.

Personal Protective Equipment (PPE)

Applicators and other handlers must wear:

- Long sleeved shirt and long pants
- Chemical-resistant gloves, including barrier laminate or Viton > 14 mils
- Protective eyewear, and
- Shoes plus socks.

Discard clothing and other absorbent materials that have been drenched or heavily contaminated with this product's concentrate. Do not reuse them. Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables exist, use detergent and hot water. Keep and wash PPE separately from other laundry.

USER SAFETY RECOMMENDATIONS

Users should:

- Remove clothing/PPE immediately if pesticide gets inside. Then wash thoroughly and put on clean clothing.
- Remove PPE immediately after handling this product. Wash the outside of gloves before removing. As soon as possible, wash thoroughly and change into clean clothing.

ENVIRONMENTAL HAZARDS

Do not apply directly to water, to areas where surface water is present, or to intertidal areas below the mean high water mark. Do not apply where runoff is likely to occur. Do not apply where weather conditions favor drift from areas treated. Do not contaminate water when disposing of equipment washwater or rinsate.

NON-TARGET ORGANISM ADVISORY

This product is toxic to plants and may adversely impact the forage and habitat of non-target organisms, including pollinators, in areas adjacent to the treated area. Protect the forage and habitat of non-target organisms by following label directions intended to minimize spray drift.

The use of this product may pose a hazard to the federally designated endangered species of Solano Grass and Wild Rice. Use of this product is prohibited in the following areas where the species are known to exist:

Solano Grass: Solano County, California: The vernal lakes area bounded by the Union Pacific Railroad and Hastings Road to the north, Highway 113 to the east, Highway 12 to the south, and Travis Air Force Base to the west.

Wild Rice: Hays County, Texas

PHYSICAL OR CHEMICAL HAZARDS

Combustible. Do not use or store near heat or open flame.

DIRECTIONS FOR USE

It is a violation of Federal law to use this product in a manner inconsistent with its labeling.

READ ENTIRE LABEL. USE STRICTLY IN ACCORDANCE WITH PRECAUTIONARY STATEMENTS AND DIRECTIONS, AND WITH APPLICABLE STATE AND FEDERAL REGULATIONS.

Do not apply this product in a way that will contact workers or other persons, either directly or through drift. Only protected handlers may be in the area during application. For any requirements specific to your State or Tribe, consult the agency responsible for pesticide regulation.

AGRICULTURAL USE REQUIREMENTS

Use this product only in accordance with its labeling and with the Worker Protection Standard, 40 CFR Part 170. This Standard contains requirements for the protection of agricultural workers on farms, forests, nurseries, and greenhouses, and handlers of agricultural pesticides. It contains requirements for training, decontamination, notification, and emergency assistance. It also contains specific instructions and exceptions pertaining to the statements on this label about personal protective equipment (PPE) and restricted-entry interval. The requirements in this box only apply to uses of this product that are covered by the Worker Protection Standard.

DO NOT enter or allow worker entry into treated areas during the restricted entry interval (REI) of 24 hours.

PPE required for early entry to treated areas that is permitted under the Worker Protection Standard and that involves contact with anything that has been treated, including plants, soil, or water, is:

- Coveralls,
- Chemical-resistant gloves, including barrier laminate or Viton > 14 mils,
- Protective eyewear, and
- Shoes plus socks

NON-AGRICULTURAL USE REQUIREMENTS

The requirements in this box apply to uses of this product that are not within the scope of the Worker Protection Standard for agricultural pesticides (40 CFR Part 170). The WPS applies when this product is used to produce agricultural plants on farms, forests, nurseries, or greenhouses.

Keep unprotected persons out of operating areas, or vicinity where there may be drift. **DO NOT** enter treated areas without protective clothing until sprays have dried.

RESISTANCE MANAGEMENT RECOMMENDATIONS

For resistance management, **Clethodim 2E** is a Group 1 herbicide. Any weed population may contain or develop plants naturally resistant to **Clethodim 2E** and other Group 1 herbicides. The resistant biotypes may dominate the weed population if these herbicides are used repeatedly in the same field. Appropriate resistance-management strategies should be followed.

To delay herbicide resistance take one or more of the following steps:

- Rotate the use of **Clethodim 2E** or other Group 1 herbicides within a growing season sequence or among growing seasons with different herbicide groups that control the same weeds in a field.
- Use tank mixtures with herbicides from a different group if such use is permitted; where information on resistance in target weed species is available, use the less resistance-prone partner at a rate that will control the target weed(s) equally as well as the more resistance-prone partner. Consult your local extension service or certified crop advisor if you are unsure as to which active ingredient is currently less prone to resistance.
- Adopt an integrated weed-management program for herbicide use that includes scouting and uses historical information related to herbicide use and crop rotation, and that considers tillage (or other mechanical control methods), cultural (e.g., higher crop seeding rates; precision fertilizer application method and timing to favor the crop and not the weeds), biological (weed-competitive crops or varieties) and other management practices.

- Scout before and after herbicide application to monitor weed populations for early signs of resistance development. Indicators of possible herbicide resistance include: (1) failure to control a weed species normally controlled by the herbicide at the dose applied, especially if control is achieved on adjacent weeds; (2) a spreading patch of non-controlled plants of a particular weed species; (3) surviving plants mixed with controlled individuals of the same species. If resistance is suspected, prevent weed seed production in the affected area by an alternative herbicide from a different group or by a mechanical method including hoeing or tillage. Prevent movement of resistant weed seeds to other fields by cleaning harvesting and tillage equipment when moving between fields, and planting clean seed.
- If a weed pest population continues to progress after treatment with this product, discontinue use of this product, and switch to another management strategy or herbicide with a different mode of action, if available.
- Contact your local extension specialist or certified crop advisors for additional pesticide resistance-management and/or integrated weed-management recommendations for specific crops and weed biotypes.

For further information or to report suspected resistance, contact Tenkoz, Inc.

TANK MIXES

Notice: Tank mixing or use of this product with any other product which is not specifically and expressly authorized by the labels shall be the exclusive risk of user, applicator, and/or application advisor. Read and follow the entire label of each product to be used in the tank mix with this product.

It is the pesticide user's responsibility to ensure that all products are registered for the intended use. Read and follow the applicable restrictions and limitations and directions for use on all product labels involved in tank mixing. Users must follow the most restrictive directions for use and precautionary statements of each product in the tank mixture.

PRODUCT INFORMATION

USE RESTRICTIONS

- Application in Long Island, New York, is restricted to no more than 16 fl oz of **Clethodim 2E (0.25 lb a.i.)** per acre per year.
- Maximum annual application rate is 1.04 lb ai/A of Clethodim containing products.
- Maximum annual applications is 4 applications per acre.
- **DO NOT** apply if rain is expected within 1 hour of application as control may be unsatisfactory.
- **DO NOT** apply a postemergence broadleaf herbicide within one day following application of **Clethodim 2E** or reduced grass control may result.
- **DO NOT** apply under conditions of stress. Applying **Clethodim 2E** under conditions that do not promote active grass growth will reduce herbicide effectiveness. These conditions include drought, excessive water, extremes in temperature, low humidity and grasses either partially controlled or stunted from prior pesticide applications. Grasses under these kinds of stressful conditions will not absorb and translocate **Clethodim 2E** effectively, and will be less susceptible to herbicide activity.
- **DO NOT** allow this product to come in contact with desirable grass crops including corn, rice, sorghum, small grains, or turf, as these and other grass crops will be injured or killed. Minor leaf spotting may occur on treated plants under certain environmental conditions. New foliage is not affected.
- Aerial applications for all tree fruits and tree nuts uses are prohibited.
- **DO NOT apply Clethodim 2E by chemigation in the states of Idaho, Montana, Oregon and Washington.**

ROTATIONAL RESTRICTIONS

DO NOT plant rotational crops until 30 days after application of **Clethodim 2E** unless the crop is listed on the **Clethodim 2E** label.

PRECAUTIONS

- Grass crops including corn, rice, sorghum, small grains, or turf, etc. are highly sensitive to **Clethodim 2E**.
- **Clethodim 2E** may not be used on vegetable crops being grown for seed production unless specific use directions are provided.
- **Clethodim 2E** is a selective post-emergence herbicide for control of annual and perennial grasses.
- **Clethodim 2E** does not control sedges or broadleaf weeds.
- Repeated use of this product (or similar post-emergence grass herbicide with the same mode of action) may lead to the selection of naturally occurring biotypes that are resistant to these products in some grass species.

- Optimal perennial grass control can be obtained if rhizomes or stolons are cut up by preplant tillage practices, (discing, plowing, etc.) to stimulate maximum emergence of grass shoots. Cultural practices, including continuous no-tillage in which the perennial grass rhizomes or stolons are not cut up, result in a very staggered, non-uniform weed emergence. Due to this non-uniform weed emergence, no fewer than 2 **Clethodim 2E** applications per year are advised at the appropriate weed-growth stage rate under continuous no-till conditions.
- While all the vegetable crops on this label have been tested and are tolerant to **Clethodim 2E**, not all specialty varieties of these crops have been tested. It is advised that, before applying **Clethodim 2E** to specialty varieties of vegetable crops on this label, crop tolerance be investigated first using a small section of the field. It is possible that injury symptoms can occur. Symptoms may appear as leaf speckling or stunting.
- It is the pesticide user's responsibility to ensure that all products are registered for the intended use. Read and follow the applicable restrictions and limitations and directions for use on all product labels involved in tank mixing. Users must follow the most restrictive directions for use and precautionary statements of each product in the tank mixture, including all crop rotational and other crop restrictions.
- Tank mixes of **Clethodim 2E** and broadleaf herbicides may result in reduced grass control. If grass regrowth occurs, an additional application of **Clethodim 2E** may be necessary.
- Repeated use of **Clethodim 2E** (or similar postemergence grass herbicides with the same mode of action) may lead to the selection of naturally occurring biotypes that are resistant to these products in some grass species.
- If poor performance occurs and cannot be attributed to adverse weather or application conditions, a resistant biotype may be present. This is most likely to occur in fields where other control strategies including crop rotation, mechanical removal, and other classes of herbicides are not used from year to year.

APPLICATION INFORMATION

Timing of Applications

Apply this product post-emergence to actively growing grasses according to rate table directions. Applications made to grass plants stressed by insufficient moisture or cold temperatures or to grass plants exceeding specified growth stages may result in unsatisfactory control. **DO NOT** apply under these conditions.

In arid regions where irrigation is used to supplement limited rainfall, this product must be applied as soon as possible after irrigation (within 7 days). A second application of this product will generally provide more effective control of perennial grass weeds than a single application. Make a second application to actively growing grass 2 to 3 weeks after emergence of new growth.

Cultivation of treated grasses 7 days prior to or within 7 days after application of this product may reduce weed control. **DO NOT apply** this product if rainfall is expected within one hour since control may be reduced.

Control Symptoms

Treated grass weeds show a reduction in vigor and growth. Early chlorosis/necrosis of younger plant tissue is followed by a progressive collapse of the remaining foliage. Symptoms will generally be observed in 7 to 14 days depending on grass species treated and environmental conditions.

ADJUVANT OR CROP OIL CONCENTRATE DIRECTIONS

Alfalfa, Cotton, Bean (dry) & Pea (shelled), Edible Podded Legume Vegetables, Peanuts (including perennial), Potato, Soybean, Bean & Pea (succulent shelled), Sugar Beet and Sunflower: Always use a crop oil concentrate at 1.0 qt./A by ground or 1 % v/v (but not less than 1 pt./A) in the finished spray volume by air. 1 to 2 qts./A of liquid fertilizer (10-34-0, 28%N or 32%N), or an equivalent amount (2.5 to 4.0 lbs./A) of spray grade ammonium sulfate (AMS) may be added to **Clethodim 2E** applications, in addition to the specified rate of crop oil concentrate. The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals, and volunteer corn.

Asparagus, Berry (low-growing), Canola, Carrot, Clover, Cranberry, Cucurbits, Flax, Fruiting Vegetables (except Okra and Tomato), Garden Beet, Garlic, Head & Stem Brassica Vegetables, Herbs, Hops, Leaf Petioles, Leafy Brassica Greens, Leafy Greens, Peppermint and Spearmint Tops, Mustard Seed, Onion (Dry Bulb), Green Onion, Pome Fruit, Root Vegetables, Safflower, Sesame, Stalk and Stem Vegetable, Stone Fruit, Strawberry, Stevia (dried leaves), Sweet Potato (Yam & other Tuberous and Corm Vegetables except Potato), Tomato, Tree Nuts and Watercress: Always use a crop oil concentrate* at 1 % v/v in the finished spray volume unless tank mix instructions indicate otherwise. Addition of liquid fertilizer is not advised for these crops.

Non-Bearing Fruit and Nut Crops, Ornamental Plants: Add a non-ionic surfactant containing at least 80% active ingredient at the rate of 1 pt. per 50 gals. (0.25% v/v). Use of crop oil concentrate is not advised since it may injure flowers and foliage.

Conifer Trees, Fallow Land (and other non-producing agricultural areas), and Non-Crop or Non-Planted Areas: Always use a crop oil concentrate* containing at least 15% emulsifier at 1% v/v (but not less than 1 pt./A) in the finished spray volume.

*Acceptable crop oil concentrates would be those that contain a minimum of 80% oils and 15% emulsifier. A crop oil concentrate must contain either a petroleum or vegetable oil base and must meet all the following criteria: be non-phytotoxic, contain only EPA-exempt ingredients, provide good mixing quality, and be successful in local experience. Highly refined vegetable oils have proven more satisfactory than unrefined vegetable oils.

Ground Application

Use of sufficient spray volumes and pressure is essential to ensure complete coverage. Use a minimum of 5 gals. and a maximum of 40 gals. of spray solution per acre. Under the following conditions a minimum of 10 gals. per acre is required: ultra narrow row cotton, narrow row soybeans, broad leaf herbicide tank mixes, perennial grasses, volunteer corn, drought or stress conditions, heavy grass pressure or when grasses are at or near maximum height. Failure to use a minimum of 10 gals. per acre under these conditions can result in poor coverage and reduced grass control requiring repeat applications. Spray pressures must reflect a minimum of 30 psi and a maximum of 60 psi at the nozzle. **DO NOT** use flood nozzles.

Applications to onions (dry bulbs and green), garlic, and shallots (dry bulbs and green) must be made in a minimum of 20 gals. of spray solution per acre.

Air Application

Use a minimum of 3 gals. of spray solution per acre unless otherwise directed in this label. Increase spray volumes up to 10 gals. as grass or crop foliage becomes dense.

For onions (dry bulb), green onion, or garlic: When applying by air **DO NOT** exceed 8 fl. oz./A (0.125 lb a.i./A) in a single application. In California, air applications to onions, garlic or shallots must be made in a minimum of 20 gals. of spray solution per acre. In states other than California, air application to onions or garlic must be made in a minimum of 10 gals. of spray solution per acre.

NOTE: Crop injury may occur when Clethodim 2E is applied to onions, garlic or shallots with aerial equipment.

Spot Treatment

When using hand sprayers or high volume sprayers utilizing hand guns, mix 1/4% to 1/2% (0.33 oz. to 0.65 oz. per gal.) **Clethodim 2E** and treat to wet vegetation, while not allowing runoff of spray solution. For uses requiring crop oil concentrate, include crop oil concentrate at 1% (1.3 oz. per gal.) by volume. For uses requiring non-ionic surfactant, include non-ionic surfactant at 1/4% (0.33 oz. per gal.) by volume.

NOTE: If Clethodim 2E is applied as a spot treatment, care must be taken to not exceed the maximum rate allowed on a "per acre" basis or crop injury may occur.

CHEMIGATION - ONION (Dry Bulb), GREEN ONION AND GARLIC SPRINKLER IRRIGATION APPLICATION

- **DO NOT** apply **Clethodim 2E** by chemigation in the states of Idaho, Montana, Oregon and Washington.

Apply **Clethodim 2E** at the high rate specified for annual grasses (16 fl. oz./A) (0.25 lb a.i./A) when the grass height is at the low end of the range (application to larger grasses may not provide adequate control). Add a crop oil concentrate containing at least 15% emulsifier at 1 quart per acre.

Apply **Clethodim 2E** in 0.1 to 0.2 acre-inch of water either at the end of a regular irrigation set or as a separate application not associated with a regular irrigation using the least amount of water that provides proper distribution and coverage. Application of more than label specified quantities of irrigation water per acre may result in decreased product performance by removing the chemical from the zone of effectiveness. Use a metering device to inject the **Clethodim 2E** into the irrigation water at a constant flow. Constant agitation must be maintained in the chemical supply tank during the entire

period of herbicide application. Inject the product with a positive displacement pump into the main line ahead of a right angle turn to ensure adequate mixing. Allow time for all lines to flush the herbicide through all nozzles before turning off irrigation water. To ensure the lines are flushed and free of remaining herbicide, a dye indicator may be injected into the lines to mark the end of the application period.

It is not advised that **Clethodim 2E** be applied through an irrigation system connected to a public water system. Public water system means a system for the provision to the public of piped water for human consumption, if such system has at least 15 service connections or regularly serves an average of at least 25 individuals daily at least 60 days out of the year.

RESTRICTIONS

- Apply this product only through sprinkler irrigation systems including center pivot, lateral move, end tow, side (wheel) roll, travelers, big gun, solid set, or hand move. **DO NOT** apply this product through any other type of irrigation system.
- **DO NOT** connect an irrigation system (including greenhouse systems) used for pesticide application to a public water system unless the label-prescribed safety devices for public water supplies are in place.
- A person knowledgeable of the chemigation system and responsible for its operation or under supervision of the responsible person, shall shut the system down and make necessary adjustments should the need arise.
- The system must contain a functional check valve, vacuum relief valve, and low pressure drain appropriately located on the irrigation pipeline to prevent water source contamination from backflow.
- The pesticide injection pipeline must contain a functional, automatic, quick-closing check valve to prevent the flow of fluid back toward the injection pump.
- The pesticide injection pipeline must also contain a functional, normally closed solenoid-operated valve located on the intake side of the injection pump and connected to the system interlock to prevent fluid from being withdrawn from the supply tank when the irrigation system is either automatically or manually shut down.
- The system must contain functional interlocking controls to automatically shut off the pesticide injection pump when the water pump motor stops.
- The irrigation line or water pump must include a functional pressure switch which will stop the water pump motor when the water pressure decreases to the point where pesticide distribution is adversely affected.
- Systems must use a metering pump, including a positive displacement injection pump (e.g., diaphragm pump) effectively designed and constructed of materials that are compatible with pesticides and capable of being fitted with a system interlock.
- **DO NOT** apply when wind speed favors drift beyond the area intended for treatment.

PRECAUTIONS

- Crop injury, lack of effectiveness, or illegal pesticide residues in the crop may result from non-uniform distribution of treated water.

If you have any questions about calibration, you should contact State Extension Service specialists, equipment manufacturers or other experts.

MANDATORY SPRAY DRIFT

Aerial Applications

- **DO NOT** release spray at a height greater than 10 ft above the vegetative canopy, unless a greater application height is necessary for pilot safety.
- For all applications, applicators are required to use a coarse or coarser spray droplet size (ASABE S572.1).
- The boom length must not exceed 65% of the wingspan for airplanes or 75% of the rotor blade diameter for helicopters.
- Applicators must use ½ swath displacement upwind at the downwind edge of the field.
- Nozzles must be oriented so the spray is directed toward the back of the aircraft.
- **DO NOT** apply when wind speeds exceed 10 miles per hour at the application site.
- **DO NOT** apply during temperature inversions.

Ground Boom Applications

- Apply with the nozzle height recommended by the manufacturer, but no more than 3 feet above the ground or crop canopy.
- For all applications, applicators are required to use a medium or coarser spray droplet size (ASABE S572.1).
- **DO NOT** apply when wind speeds exceed 10 miles per hour at the application site.
- **DO NOT** apply during temperature inversions.
- For all other ground applications, the nozzle must be no more than 3 feet from the target vegetation.

SPRAY DRIFT ADVISORIES

THE APPLICATOR IS RESPONSIBLE FOR AVOIDING OFF-SITE SPRAY DRIFT.

BE AWARE OF NEARBY NON-TARGET SITES AND ENVIRONMENTAL CONDITIONS.

• **IMPORTANCE OF DROPLET SIZE**

An effective way to reduce spray drift is to apply large droplets. Use the largest droplets that provide target pest control. While applying larger droplets will reduce spray drift, the potential for drift will be greater if applications are made improperly or under unfavorable environmental conditions.

Controlling Droplet Size - Ground Boom

- Volume - Increasing the spray volume so that larger droplets are produced will reduce spray drift. Use the highest practical spray volume for the application. If a greater spray volume is needed, consider using a nozzle with a higher flow rate.
- Pressure - Use the lowest spray pressure recommended for the nozzle to produce the target spray volume and droplet size.
- Spray Nozzle - Use a spray nozzle that is designed for the intended application. Consider using nozzles designed to reduce drift.

Controlling Droplet Size - Aircraft

- Adjust Nozzles - Follow nozzle manufacturers recommendations for setting up nozzles. Generally, to reduce fine droplets, nozzles should be oriented parallel to the airflow in flight.

• **BOOM HEIGHT - Ground Boom**

Use the lowest boom height that is compatible with the spray nozzles that will provide uniform coverage. For ground equipment, the boom should remain level with the crop and have minimal bounce.

• **RELEASE HEIGHT - Aircraft**

Higher release heights increase the potential for spray drift. When applying aerially to crops, **DO NOT** release spray at a height greater than 10 ft above the crop canopy, unless a greater application height is necessary for pilot safety.

• **Handheld Technology Applications:**

Take precautions to minimize spray drift.

• **SHIELDED SPRAYERS**

Shielding the boom or individual nozzles can reduce spray drift. Consider using shielded sprayers. Verify that the shields are not interfering with the uniform deposition of the spray on the target area.

• **TEMPERATURE AND HUMIDITY**

When making applications in hot and dry conditions, use larger droplets to reduce effects of evaporation.

• **TEMPERATURE INVERSIONS**

Drift potential is high during a temperature inversion. Temperature inversions are characterized by increasing temperature with altitude and are common on nights with limited cloud cover and light to no wind. The presence of an inversion can be indicated by ground fog or

by the movement of smoke from a ground source or an aircraft smoke generator. Smoke that layers and moves laterally in a concentrated cloud (under low wind conditions) indicates an inversion, while smoke that moves upward and rapidly dissipates indicates good vertical air mixing. Avoid applications during temperature inversions.

- **WIND**

Drift potential generally increases with wind speed. AVOID APPLICATIONS DURING GUSTY WIND CONDITIONS.

Applicators need to be familiar with local wind patterns and terrain that could affect spray drift.

- **Boom-less Ground Applications:**

Setting nozzles at the lowest effective height will help to reduce the potential for spray drift.

CROP SPECIFIC DIRECTIONS AND RESTRICTIONS FOR CLETHODIM 2E

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Alfalfa including: Sainfoin Holy Clover Birdsfoot trefoil ⁽³⁾	15 days before grazing, feeding or harvesting (cutting) for forage or hay	6-16 fl. oz. ⁽⁴⁾ [0.09375 - 0.25 lb a.i.]	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁵	For repeat applications make on a minimum of a 14-day interval. Refer to tank mix partners for feeding, grazing and harvesting restrictions. ^(5,6) The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. RESTRICTIONS: • DO NOT apply more than 16 fl. oz./A (0.25 lb ai) in a single application. • DO NOT make more than 2 applications per acre per year. • DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Bean, Dry except Soybean [Dried Shelled Pea and Bean (except soybean)] [Subgroup 6C] including: Bean (<i>Lupinus</i> spp.) Grain Sweet White White Sweet Bean (<i>Phaseolus</i> spp.) Field Kidney Lima (dry) Navy Pinto Tepary Bean (<i>Vigna</i> spp.) Adzuki bean Black-eyed pea Catjang Cowpea Crowder pea Moth bean Mung bean Rice bean Southern pea Urd bean Broad bean (dry) Chickpea (garbanzo) Guar Lablab bean Lentil	30 days	6-16 fl. oz. [0.09375 - 0.25 lb a.i.]	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air. ⁽⁵⁾	For repeat applications make on a minimum of a 14-day interval. Refer to appropriate Table for reduced rate directions for the control of small annual grasses. The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application.] DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. - For reduced rate control for small annual grasses: DO NOT make more than 4 applications at 4 fl oz (0.0625 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Bean, Succulent Shelled [Succulent Shelled Pea and Bean Subgroup 6B] including: Bean (Phaseolus spp.) Lima bean (green) Broad bean [succulent] Bean (Vigna spp.) Black-eyed pea Cowpea Southern pea	21 days	6-8 fl. oz. [0.09375 - 0.125 lb a.i.]	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁵	Refer to appropriate Table for reduced rate directions for the control of small annual grasses. The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A [0.125 lb a.i./A] in a single application. DO NOT apply more than one 1 application per acre per year. DO NOT apply more than 8 fl. oz. [0.125 lb a.i.] per acre per year.
Beet, Garden	30 days	6-8 fl. oz. [0.09375 - 0.125 lb a.i.]	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A [0.125 lb a.i./A] in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz [0.500 lb ai] per acre per year.
Berry Low Growing (except Cranberry and Strawberry) Subgroup 13-07G including: Bearberry Bilberry Blueberry, lowbush Cloudberry Lingonberry Muntries Partridgeberry	45 days	4-8 fl. oz. [0.0625 - 0.125 lb a.i.]	Non-Ionic surfactant (NIS) at 0.25% v/v	For repeat applications make on a minimum of a 14 day interval. Verify crop safety to Clethodim 2E on a small area of the crop, at the desired Clethodim 2E rate and with the same non-ionic surfactant (NIS) that will be used on the field. If no crop response is evident 7 days after treatment, Clethodim 2E may be used on the entire field at the rate tested and with the same NIS used in the crop safety test. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A [0.125 lb a.i./A] in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz [0.500 lb ai] per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Brassica Head and Stem Vegetable [Crop Group 5-16]* including: Broccoli Brussels sprouts Cabbage Cabbage, Chinese (napa) Cauliflower cultivars, varieties and/or hybrids of these commodities	30 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. *Not Registered for Use in California
Brassica Leafy Greens [Subgroup 4-16B]* (except Radish Leaves, Turnip Greens and Watercress) including: Arugula Broccoli, Chinese Broccoli Raab Cabbage, Abyssinian Cabbage, Chinese (bok choy) Cabbage, Seakale Collards Cress, Garden Cress, Upland Hanover Salad Kale Maca (leaves) Mizuna Mustard greens Mustard spinach Rape greens Rocket, Wild Shepherd's Purse cultivars, varieties, and/or hybrids of these commodities	14 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. *Not Registered for Use in California

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Canola [including Rapeseed Subgroup 20A* except flax seed, mustard seed and sesame seed] Borage Crambe Cuphea Echium Gold of Pleasure (Camelina) Hare's Ear Mustard Lesquerella Lunaria Meadowfoam Milkweed Oil Radish Poppy seed Rapeseed (canola) Sweet Rocket	70 days	4-6 fl. oz. (0.0625 – 0.09375 lb a.i.)	1.0% v/v in the finished spray volume	Verify crop safety to Clethodim 2E on a small area of the crop, at the desired Clethodim 2E rate and with the same non-ionic surfactant (NIS) that will be used on the field. If no crop response is evident 7 days after treatment, Clethodim 2E may be used on the entire field at the rate tested and with the same NIS used in the crop safety test. RESTRICTIONS: DO NOT apply more than 6 fl. oz./A (0.09375 lb a.i./A) in a single application. DO NOT make more than 1 application per acre per year. DO NOT apply more than 6 fl. oz./A (0.09375 lb a.i./A) in a year. DO NOT apply after crop has begun bolting. Crop injury may occur when this product is applied during the bloom period *Not Registered for Use in California
Carrot	30 days	6-8 fl. oz. (0.09375 – 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Clover (Idaho, Oregon and Washington only)	15 days before grazing, feeding or harvesting (cutting) for forage or hay	6-16 fl. oz. (0.09375 – 0.25 lb a.i.)	1.0% v/v in the finished spray volume	RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 1 application per acre per year. DO NOT apply more than 16 fl oz (0.25 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Cotton (including cotton grown for seed)	60 days	6-16 fl. oz. [0.09375 - 0.25 lb a.i.]	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁵	The addition of Ammonium Sulfate (AMS) has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. DO NOT graze treated field or feed treated forage or hay to livestock.
Cranberry	30 days	6-8 fl. oz. [0.09375 - 0.125 lb a.i.]	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. DO NOT apply between the "hook" stage and full fruit set.
Conifer Trees Fallow Land (and other non-producing agricultural areas) Non-Crop or Non-Planted Areas⁷	N/A	6-16 fl. oz. [0.09375 - 0.25 lb a.i.]	1.0% v/v (but not less than 1 pt./A) in the finished spray volume using a crop oil concentrate containing at least 15% emulsifier.	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. DO NOT plant any crop for 30 days after application unless clethodim is registered for use in that crop.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Flax*	60 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	Apply prior to bloom. Crop injury may result when this product is applied during the bloom period. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 16 fl. oz. (0.25 lb a.i.) per acre per year. - For reduced rate control for small annual grasses: DO NOT make more than 4 applications at 4 fl. oz. (0.0625 lb ai) per acre per year. *Not Registered for Use in California
Fruiting Vegetables (except Okra and Tomato) [Crop Group 8-10] including: African Eggplant Bush Tomato Bell Pepper Cocona Currant Tomato Eggplant Garden Huckleberry Goji Berry Groundcherry Martynia Naranjilla Pea Eggplant Pepino Nonbell Pepper Roselle Scarlet Eggplant Sunberry Peppers (all) Tomatillo Tree Tomato Cultivars, varieties, and/or hybrids of these.	20 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
[Subgroup 19A] Herbs including: Angelica Balm Basil Borage Burnet Chamomile Catnip Chervil (dried) Chive Chinese chive Clary Coriander (leaf) Costmary Culantro (leaf) Curry (leaf) Dillweed Horehound Hyssop Lemongrass Lovage (leaf) Marigold Marjoram Nasturtium Parsley (dried) Pennyroyal Rosemary Rue Sage Savory, Summer and Winter Sweet Bay Tansy Tarragon Thyme Wintergreen Woodruff Wormwood	14 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	This product has not been tested on all herbs and herb varieties. It is the responsibility of the user to test this product on a small portion of the crop to be treated before treating the entire field. Crop resistance to this product must be verified on a small area of the herb crop at the desired rate and with the same crop oil concentrate that will be used on the herb field. If no crop response is evident seven (7) days after treatment, this product may be used on the entire field at the same rate tested and with the same crop oil used in the resistance test. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: • DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. • DO NOT make more than 4 applications per acre per year. • DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Hops	21 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
[Subgroup 22B] Leaf Petiole Vegetables* including: Cardoon Celery Chinese celery Fuki Rhubarb Udo Zuiki	30 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. *Not for Use in California

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
[Subgroup 4-16A] Leafy Greens* including: Amaranth, Chinese Amaranth, Leafy Aster, Indian Blackjack Cat's Whiskers Cham-chwi Cham-na-mul Chervil (fresh leaves) Chipilin Chrysanthemum, Garland Cilantro (fresh leaves) Corn salad Cosmos Dandelion (leaves) Dang-gwi (leaves) Dillweed Dock Dol-nam-mul Ebolo Endive Escarole Fameflower Feather Cockscomb Good King Henry Huauzontle Jute (leaves) Lettuce, Bitter Lettuce, Head Lettuce, Leaf Orach Parsley (fresh leaves) Plantain, Buckhorn Primrose, English Purslane, Garden Purslane, Winter Radicchio Spinach Spinach, Malabar Spinach, New Zealand Spinach (Tanier) Swiss Chard Violet, Chinese (leaves) Cultivars, varieties, and/or hybrids of these commodities.	14 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. *Not Registered for Use in California

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
[Subgroup 6A] Legume Vegetables, Edible Podded including: Bean (Phaseolus spp.) runner snap wax Bean (Vigna spp.) asparagus Chinese longbean moth yardlong jackbean Pea (Pisum spp.) dwarf edible-podded snow sugar snap pigeon Soybean (immature seed) Sword bean	21 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁵	For peas, apply before bloom, but no later than 21 days before harvest. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 1 application per acre per year. DO NOT apply more than 8 fl oz (0.125 lb ai) per acre per year.
Melon [Subgroup 9A] Citron melon Muskmelon (including cantaloupe) Watermelon	14 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray	For repeat applications make on a minimum of a 14 day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Peppermint and Spearmint Tops	21 days	6-16 fl. oz. ⁴ (0.09375 - 0.25 lb a.i.)	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Mustard Seed*	75 days	4-6 fl. oz. (0.0625 – 0.09375 lb a.i.)	1.0% v/v in the finished spray volume	Crop injury may occur when this product is applied during the bloom period. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: • DO NOT apply more than 6 fl. oz./A (0.09375 lb a.i./A) in a single application. • DO NOT apply more than 12 fl. oz. (0.188 lb a.i.) per acre per year. • DO NOT apply after crop has begun bolting. • DO NOT make more than 2 applications per acre per year. • For Reduced Rate Control for Small Annual Grasses: DO NOT make more than 3 applications at 4 fl oz (0.0625 lb ai) per acre per year. *Not Registered for Use in California
Okra*	3 days	6-8 fl. oz. (0.09375 – 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: • DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. • DO NOT make more than 4 applications per acre per year. • DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. *Not Registered for Use in California

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
[Bulb Onion] [Subgroup 3-07A] Onion (Dry Bulb Only) Including Daylily, Bulb Fritillaria, Bulb Garlic, Bulb Garlic, Great-headed, Bulb Garlic, Serpent, Bulb Lily, Bulb Onion, Bulb Onion, Chinese, Bulb Onion, Pearl Onion, Potato, Bulb Shallot, Bulb Cultivars, varieties, and/or hybrids of these.	45 days	6-16 fl. oz. [0.09375 – 0.25 lb a.i.]	1.0% v/v in the finished spray volume	Minimum of 20 gals./A spray volume by ground in entire US. Minimum of 20 gals./A spray volume by air in California. In states other than California, air applications to onions, garlic or shallots must be made in a minimum of 10 gals./A. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: • DO NOT apply more than 16 fl. oz./A [0.25 lb a.i./A] in a single application. • DO NOT make more than 2 applications per acre per year. • DO NOT apply more than 32 fl oz [0.500 lb ai] per acre per year. • When applying by air, DO NOT exceed 8 fl oz [0.125 lb ai] per acre per application and 16 fl oz [0.25 lb ai] per acre per year. • If Clethodim 2E is applied as a spot treatment to onion or garlic, DO NOT exceed the maximum rate allowed on a "per acre" basis (16 fl. oz. [0.25 lb ai] per acre). • In California, DO NOT apply Clethodim 2E to garlic or onion until the crop has at least two full leaves. Use a 14 day spray interval between the application of Clethodim 2E and liquid nitrogen or other herbicide applications. Injury to crop may occur when shorter intervals are observed. For Garlic: • When applying by ground, DO NOT exceed 8 fl oz [0.125 lb ai] per acre per application or 16 fl oz [0.250 lb ai] per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Onion, Green [Subgroup 3-07B]* including: Chive (fresh leaves) Chive, Chinese (fresh leaves) Elegans Hosta Fritillaria (leaves) Kurrat Lady's Leek Leek Leek, Wild Onion, Beltsville (bunching) Onion (fresh) Onion, Green Onion, Macrostem Onion, Tree (tops) Onion, Welsh (tops) Shallot (fresh leaves) Cultivars, varieties, and/or hybrids of these commodities.	14 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. - For air applications to onion, DO NOT exceed 8 fl oz (0.125 lb ai) per acre per application. - In California for air applications to onion, DO NOT exceed 2 applications per year. - If Clethodim 2E is applied as a spot treatment to onion or garlic, DO NOT exceed the maximum rate allowed on a "per acre" basis (8 fl. oz. (0.125 lb ai) per acre). - In California, DO NOT apply Clethodim 2E to onion or garlic until the crop has at least two full leaves. Use a 14-day spray interval between the application of Clethodim 2E and liquid nitrogen or other herbicide applications. Injury to crop may occur when shorter intervals are observed. *Not Registered for Use in California
Peanut (including perennial)	40 days	6-16 fl. oz. (0.09375 - 0.25 lb a.i.)	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ^{12, 31}	The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
[Subgroup 6C] Pea, Dry Shelled including: Pea (<i>Pisum</i> spp.) Field Pigeon	30 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁽⁵⁾	For repeat applications make on a minimum of a 14 day interval. Apply before bloom but not later than 30 days prior to harvest. Refer to appropriate Table for reduced rate directions for the control of small annual grasses. The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. Applications of Clethodim 2E to peas during the bloom period could result in severe crop injury, including loss of yield and delayed maturity. RESTRICTIONS: • DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. • DO NOT make more than one 1 application per acre per year. • DO NOT apply more than 8 fl oz (0.125 lb ai) per acre per year. • For Reduced Rate Control for Small Annual Grasses: DO NOT make more than 2 applications at 4 fl oz (0.0625 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
[Subgroup 6B] Pea, Succulent Shelled including: Pea (<i>Pisum</i> spp.) English pea Garden pea Green pea Pigeon pea	21 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁽⁵⁾	Apply before bloom but not later than 21 days prior to harvest. Refer to appropriate Table for reduced rate directions for the control of small annual grasses. The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: • DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. • DO NOT make more than one 1 application per acre per year. • DO NOT apply more than 8 fl oz (0.125 lb ai) per acre per year. • For Reduced Rate Control for Small Annual Grasses: DO NOT make more than 2 applications at 4 fl oz (0.0625 lb ai) per acre per year.
[Crop Group11-10] Pome Fruit including Apple Azarole Crabapple Loquat Mayhaw Medlar Pear Pear, Asian Quince Quince, Chinese Quince, Japanese Tejocote	14 days	4-8 fl. oz. (0.0625 - 0.125 lb a.i.)	Non-ionic surfactant (NIS) at 0.25% v/v	For repeat applications make on a minimum of a 14 day interval. RESTRICTIONS: • DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. • DO NOT make more than 4 applications per acre per year. • DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Potato	30 days	6-16 fl. oz. (0.09375 - 0.25 lb a.i.)	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁽⁵⁾	The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Radish Radish Leaves	15 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application DO NOT make more than 2 applications per acre per year. DO NOT apply more than 16 fl. oz./A (0.25 lb. ai/A) per acre per year.
[Subgroup 1B] Root Vegetables (except Sugar Beet and Radish), including: Burdock, Edible Celeriac Chervil, Turnip Rooted Chicory Ginseng Horseradish Parsley, Turnip Rooted Parsnip Radish, Oriental Rutabaga Salsify Salsify, Black Salsify, Spanish Skirret Turnip	30 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Safflower	70 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Sesame	14 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. DO NOT apply during flowering.
Soybean	60 days	6-16 fl. oz. (0.09375 - 0.25 lb a.i.)	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁽⁵⁾	Refer to appropriate Table for reduced rate directions for the control of small annual grasses. The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals, and volunteer corn. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. DO NOT graze treated fields or feed treated forage or hay to livestock. - For Reduced Rate Control for Small Annual Grasses: DO NOT make more than 4 applications at 4 fl oz (0.0625 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Squash/Cucumber [Subgroup 9B] Chayote (fruit) Chinese waxgourd (Chinese preserving melon) Cucumber Gherkin Gourd, edible (includes hyotan, cucuzza, hechima, Chinese okra) <i>Momordica</i> spp. (includes balsam apple, balsam pear, bittermelon, Chinese cucumber) Pumpkin Squash, Summer Squash, Winter (includes butternut squash, calabaza, hubbard squash, acorn squash, spaghetti squash)	14 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Stalk and Stem Vegetable [Subgroup 22A]* Agave Aloe Vera Asparagus Bamboo Shoots Celtnce Fennel, Florence (fresh leaves and stalk) Fern, Fiddlehead (edible) Kale, Sea Kohlrabi Palm Hearts Prickly Pear (pads) Prickly Pear, Texas (pads) cultivars, varieties, and/or hybrids of these commodities	1 day	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application DO NOT make more than 2 applications per acre per year. DO NOT apply more than 16 fl. oz./A (0.25 lb. ai/A) per acre in a year. *Not Registered for Use in California

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Stevia, dried leaves	14 days	4-8 fl. oz. (0.0625 – 0.125 lb a.i.)	Non-ionic surfactant (NIS) at 0.25% v/v	For repeat applications make on a minimum of a 14 day interval. Clethodim 2E has not been tested on all varieties. It is the responsibility of the user to test Clethodim 2E on a small portion of the crop to be treated before treating the entire field. Verify crop safety to Clethodim 2E on a small area of the crop, at the desired Clethodim 2E rate and with the same non-ionic surfactant (NIS) that will be used on the field. If no crop response is evident 7 days after treatment, Clethodim 2E may be used on the entire field at the rate tested and with the same NIS used in the crop safety test. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
[Crop Group 12-12] Stone Fruit including Apricot Apricot, Japanese Capulin Cherry, Black Cherry, Nanking Cherry, Sweet Cherry, Tart Jujube, Chinese Nectarine Peach Plum Plum, American Plum, Beach Plum, Canada Plum, Cherry Plum, Chickasaw Plum, Damson Plum, Japanese Plum, Klamath Plum, Prune Plumcot Sloe	14 days	4-8 fl. oz. (0.0625 – 0.125 lb a.i.)	Non-ionic surfactant (NIS) at 0.25% v/v	For repeat applications make on a minimum of a 14 day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Strawberry	4 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Sugar Beet	40 days	6-16 fl. oz. (0.09375 - 0.25 lb a.i.)	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁽⁵⁾	Refer to appropriate Table for reduced rate directions for the control of small annual grasses. The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals, and volunteer corn. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. - For Reduced Rate Control for Small Annual Grasses: DO NOT make more than 4 applications at 4 fl oz (0.0625 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
[Subgroup 20B] Sunflower	70 days	6-16 fl. oz. [0.09375 - 0.25 lb a.i.]	1 qt. by ground or 1.0% v/v (but no less than 1 pt./A) by air ⁽⁵⁾	The addition of AMS has shown improved grass control for difficult to control species including: quackgrass, rhizome Johnsongrass, red rice, wild oats, volunteer cereals, and volunteer corn. Verify crop safety to Clethodim 2E on a small area of the crop, at the desired Clethodim 2E rate and with the same crop oil concentrate that will be used on the field. If no crop response is evident seven (7) days after treatment, Clethodim 2E may be used on the entire field at the rate tested and with the same crop oil used in the crop safety test. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: • DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. • DO NOT make more than 2 applications per acre per year. • DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. • For Reduced Rate Control for Small Annual Grasses: DO NOT make more than 4 applications at 4 fl oz (0.0625 lb ai) per acre per year.
Tomato	20 days	6-16 fl. oz. [0.09375 - 0.25 lb a.i.]	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: • DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. • DO NOT make more than 2 applications per acre per year. • DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Tree Nuts [Crop Group 14-12]* Including: African Nut-tree Almond Beechnut Brazil Nut Brazilian Pine Bunya Bur Oak Butternut Cajou Nut Candlenut Cashew Chestnut Chinquapin Coconut Coquito Nut Dika Nut Ginkgo Guiana Chestnut Hazelnut (Filbert) Heartnut Hickory Nut Japanese Horse- chestnut Macadamia Nut Mongongo Nut Monkey-pot Monkey Puzzle Nut Okari Nut Pachira Nut Peach Palm Nut Pecan Pequi Pili Nut Pine Nut Pistachio Sapucaia Nut Tropical Almond Walnut, Black Walnut, English Yellowhorn Cultivars, varieties, and/or hybrids of these.	14 days	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. Aerial applications for all tree nuts are prohibited. *Not Registered for Use in California

Crops ¹	Minimum Time from Application to Harvest (PHI)	Use Rate Per Acre	Crop Oil Concentrate Rates Per Acre ²	Special Use Instructions
Tuberous and Corm Vegetables Subgroup [Subgroup 1C] (except Potato) Including Sweet Potato Yam Artichoke Chinese Jerusalem Cassava Bitter Sweet Ginger	30 days	6-16 fl. oz. [0.09375 - 0.25 lb a.i.]	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14 day interval. The addition of Ammonium Sulfate (AMS) has shown improved grass control for difficult to control species including: quackgrass, rhizome, Johnsongrass, red rice, wild oats, volunteer cereals and volunteer corn. RESTRICTIONS: DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Turnip Greens	14 days	6-8 fl. oz. [0.09375 - 0.125 lb a.i.]	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Watercress*	30 days	6-8 fl. oz. [0.09375 - 0.125 lb a.i.]	1.0% v/v in the finished spray volume	For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year. DO NOT apply when watercress is under flood conditions. DO NOT apply Clethodim 2E when water is in the field and hold water for at least 24 hours after an application. *Not Registered for Use in California

N/A - Not Applicable

¹ **Clethodim 2E** is not to be used on vegetable crops being grown for seed production unless specific use directions are provided.

² Acceptable crop oil concentrates would be those which contain a minimum of 80% oils and 15% emulsifier. A crop oil concentrate must contain either a petroleum or vegetable oil base and must meet all the following criteria: be non phytotoxic, contain only EPA-exempt ingredients, provide good mixing

quality and be successful in local experience. Highly refined vegetable oils have proven more satisfactory than unrefined vegetable oils. See the Addition of Adjuvant and Crop Oil Concentrate section for further information.

³ **Clethodim 2E** may be applied to seedling or established alfalfa grown for seed, hay, silage, green chop or direct grazing.

⁴ For weed control in established alfalfa and peppermint and spearmint tops, the minimum use rate is 10 fl. oz./A [0.156 lb a.i./A]

⁵ 1 to 2 qts./A of liquid fertilizer (10-34-0, 28%N or 32%N), or an equivalent amount (2.5 to 4.0 lbs./A) of spray grade ammonium sulfate (AMS) may be added to **Clethodim 2E** applications, in addition to the specified rate of crop oil concentrate.

⁶ **DO NOT** apply **Clethodim 2E** plus 2,4-DB as a tank mix to alfalfa unless the 60 day feeding, grazing, and harvesting restriction on the 2,4-DB label can be observed.

⁷ Non-Crop or Non-Planted Areas including rights-of-way, railroads, highways, roads, dividers, medians, pipelines, public utility lines, pumping stations and substations, around airports, electric utilities, commercial buildings, manufacturing plants, storage yards, rail yards, fence lines, parkways, and post-harvest croplands, also beneath greenhouse benches and around golf courses.

IMPORTANT

Plant resistance to **Clethodim 2E** at labeled rates has been found to be acceptable for the indicated genera and species listed below. Due to variability within species, crop growth stage, environmental conditions, and application techniques, it is advised that the user determine if the herbicide can be used safely on a few plants prior to widespread application. Neither the seller nor the manufacturer of **Clethodim 2E** have investigated the safety factor to plants not listed on the label.

INSTRUCTIONS FOR ANNUAL GRASSES

(EXCEPT FOR IN ESTABLISHED ALFALFA AND PEPPERMINT AND SPEARMINT TOPS)

- Apply only to actively growing grasses at specified weed heights.
- Apply when the first grass weed species in a mixed grass weed population reaches the specified growth stage for treatment.
- Use the high rate under heavy grass pressure and/or when grasses are at maximum height.

Restrictions

- **DO NOT** exceed the maximum rate per application listed in the table, Crop Specific Use Directions and Restrictions for **Clethodim 2E**.
- **DO NOT** exceed the maximum number of applications per year listed in the table, Crop Specific Use Directions and Restrictions for **Clethodim 2E**.
- **DO NOT** exceed the maximum yearly rate listed in the table, Crop Specific Use Directions and Restrictions for **Clethodim 2E**.

GRASS SPECIES	SCIENTIFIC NAME	WEED HEIGHT* (inches)	RATE FL. OZ./ACRE (lb a.i./A)	HIGH RATE ¹ FL. OZ./ACRE (lb a.i./A)
Barnyardgrass	<i>Echinochloa crus-galli</i>	2 to 8	6 [0.09375]	8 [0.125]
Broadleaf Signalgrass	<i>Brachiaria platyphylla</i>	2 to 6	6 [0.09375]	8 [0.125]
Brome				
California	<i>Bromus carinatus</i>	2 to 6	6 [0.09375]	8 [0.125]
Cheat	<i>Bromus secalinus</i>	2 to 6	6 [0.09375]	8 [0.125]
Downy	<i>Bromus tectorum</i>	2 to 6	6 [0.09375]	8 [0.125]
Ripgut	<i>Bromus diandrus</i>	2 to 6	6 [0.09375]	8 [0.125]
Canarygrass	<i>Phalaris canariensis</i>	1 to 4	6 [0.09375]	8 [0.125]
Crabgrass				
Hairy	<i>Digitaria adscendens</i>	2 to 6**	6 [0.09375]	8 [0.125]
Large	<i>Digitaria sanguinalis</i>	2 to 6**	6 [0.09375]	8 [0.125]
Smooth	<i>Digitaria ischaemum</i>	2 to 6**	6 [0.09375]	8 [0.125]
Southern	<i>Digitaria ciliaris</i>	2 to 6**	6 [0.09375]	8 [0.125]
Crowfootgrass	<i>Dactyloctenium aegyptium</i>	2 to 6**	6 [0.09375]	8 [0.125]
Fall Panicum	<i>Panicum dichotomiflorum</i>	2 to 8	6 [0.09375]	8 [0.125]
Field Sandbur	<i>Cenchrus incertus</i>	2 to 6	6 [0.09375]	8 [0.125]

GRASS SPECIES	SCIENTIFIC NAME	WEED HEIGHT* (inches)	RATE FL. OZ./ACRE (lb a.i./A)	HIGH RATE ¹ FL. OZ./ACRE (lb a.i./A)
Foxtail				
Giant	<i>Setaria faberi</i>	2 to 12	6 [0.09375]	8 [0.125]
Green	<i>Setaria viridis</i>	2 to 8	6 [0.09375]	8 [0.125]
Yellow	<i>Setaria glauc</i>	2 to 8	6 [0.09375]	8 [0.125]
Goosegrass	<i>Eleusine indica</i>	2 to 6**	6 [0.09375]	8 [0.125]
Itchgrass	<i>Rottboellia cochinchinensis</i>	2 to 6	6 [0.09375]	8 [0.125]
Junglerice	<i>Echinochloa colona</i>	2 to 6	6 [0.09375]	8 [0.125]
Lovegrass (Stinkgrass)	<i>Eragrostis cilianensis</i>	2 to 6	6 [0.09375]	8 [0.125]
Rabbitsfootgrass	<i>Polypogon monspeliensis</i>	1 to 4	6 [0.09375]	8 [0.125]
Red Rice	<i>oryza sativa</i>	1 to 3	6 [0.09375]	8 [0.125]
Ryegrass				
Hardy	<i>Lolium remotum</i>	2 to 6	6 [0.09375]	8 [0.125]
Italian	<i>Lolium multiflorum</i>	2 to 6	6 [0.09375]	8 [0.125]
Seedling Johnsongrass	<i>Sorghum halepense</i>	4 to 10	6 [0.09375]	8 [0.125]
Shattercane	<i>Sorghum bicolor</i>	6 to 18	6 [0.09375]	8 [0.125]
Southwestern Cupgrass	<i>Eriochloa gracilis</i>	2 to 6	6 [0.09375]	8 [0.125]
Sprangle top				
Amazon	<i>Leptochloa panicoides</i>	2 to 6	6 [0.09375]	8 [0.125]
Bearded	<i>Leptochloa fascicularis</i>	2 to 6	6 [0.09375]	8 [0.125]
Mexican	<i>Leptochloa uninervia</i>	2 to 6	6 [0.09375]	8 [0.125]
Red	<i>Leptochloa filiformis</i>	2 to 6	6 [0.09375]	8 [0.125]
Texas Panicum	<i>Panicum flexanum</i>	2 to 6	6 [0.09375]	8 [0.125]
Barley	<i>Hordeum vulgare</i>	2 to 6	6 [0.09375]	8 [0.125]
Oats	<i>Avena sativa</i>	2 to 6	6 [0.09375]	8 [0.125]
Rye	<i>Secale cereale</i>	2 to 6	6 [0.09375]	8 [0.125]
Wheat	<i>Triticum aestivum</i>	2 to 6	6 [0.09375]	8 [0.125]
Volunteer Corn ³	<i>Zea mays</i>	4 to 12	4 [0.0625]	6 [0.09375]
Volunteer Corn [S.R.] ⁴	<i>Zea mays</i>	4 to 12	8 (suppression only) [0.125]	
Volunteer Grain Sorghum	<i>Sorghum bicolor</i>	8 to 12	6 [0.09375]	8 [0.125]
Wild Oats	<i>Avena fatua</i>	2 to 6	6 [0.09375]	8 [0.125]
Wild Proso Millet	<i>Panicum miliaceum</i>	2 to 10	6 [0.09375]	8 [0.125]
Witchgrass	<i>Panicum capillare</i>	2 to 8	6 [0.09375]	8 [0.125]
Woolly Cupgrass	<i>Eriochloa villosa</i>	2 to 8	6 [0.09375]	8 [0.125]

*Generally occurs between 3-leaf stage and tillering.

**Length of lateral growth.

¹ Rates higher than 8 fl. oz./A [0.125 lb a.i./A] may be applied in certain geographic areas, cropping situations, or environmental conditions, where experience has shown that higher rates are needed for satisfactory control of annual grasses. In these situations, rates from 8 to 16 fl. oz./A [0.125 to 0.25 lb a.i./A] may be applied. **DO NOT** apply more than 6 fl. oz./A of **Clethodim 2E** [0.09375 lb a.i.] per application to rapeseed subgroup 20A listed crops or mustard seed.

² When a cereal grain crop (including wheat) is interseeded for crop establishment or is planted as wind breaks to aid crop establishment, the minimum **Clethodim 2E** use rate for control is 8 fl. oz./A [0.125 lb a.i./A].

³ Includes Roundup Ready®, Liberty Link® and IMI-CORN® volunteer corn.

⁴ Sethoxydim resistant volunteer corn.

INSTRUCTIONS FOR ANNUAL & PERENNIAL GRASS CONTROL IN ESTABLISHED ALFALFA AND PEPPERMINT AND SPEARMINT TOPS WITH CLETHODIM 2E

GRASS SPECIES	WEED STAGE	RATE FL. OZ./ACRE (lb a.i./A)	HIGH RATE (lb a.i./A)
Annual & Perennial Grasses Listed in Grass Table	See Instructions for Annual Grasses Table	10 [0.156]	16 [0.25]

Mowing: The best control of annual grasses can be achieved by applying **Clethodim 2E** before grass weeds are mowed. Once a grass is mowed it becomes tougher to control, as much of the available leaf surface has been removed. In areas without a killing frost, some annuals can over-winter after having been mowed multiple times. These grasses form large crowns and may contain many viable buds. These grasses, even though they may be an annual grass, may require repeated applications of **Clethodim 2E** for partial or complete control.

Irrigated Alfalfa and Peppermint and Spearmint Tops: Irrigation practices can be very critical to the successful use of **Clethodim 2E** in established alfalfa and peppermint and spearmint tops and may be necessary to initiate active growth of the weeds prior to application. Generally applications 2 to 4 days after an irrigation are most effective. Irrigation made shortly after application (2 days) can be effective, but more consistent grass control occurs when the irrigation is made before the application.

Aerial Application: Apply **Clethodim 2E** in a minimum of 10 GPA in established alfalfa and peppermint and spearmint tops when applying by air.

Annual Grass Control: Apply **Clethodim 2E** at the grass sizes indicated in the Directions for Annual Grass Table and rates indicated. If a grass has been cut, apply **Clethodim 2E** after active growth has resumed and regrowth has reached the minimum height and before it reaches the maximum height indicated. Apply before the alfalfa/peppermint and spearmint tops canopy covers the grasses and interferes with the spray coverage. Some annual grasses are spring- and summer-germinating plants, while others are fall-germinating plants, and the time they are actively growing and most susceptible to **Clethodim 2E** may vary from region to region. Also, some annuals germinate over an extended period of time, and because control of small grasses is desired, applications after each weed flush may be required. As a general rule spray spring- and summer-germinating grasses as early in the season as possible, after initial green-up. Spray fall-germinating weeds in the fall soon after they begin growing but before any damage is done due to frost. Late fall applications may be less effective due to environmental conditions, including frost, slower plant growth, or the onset of flowering.

Perennial Grass Control: **Clethodim 2E** effectively controls perennial grasses including bermudagrass, Johnsongrass, quackgrass, wirestem muhly, tall fescue, foxtail barley and orchardgrass. Due in part to lack of tillage, perennial grasses are more difficult to control in a perennial crop including established alfalfa or peppermint and spearmint tops. A program of repeated applications is usually necessary for best results. The best way to control perennial grasses is to do so in the year of stand establishment before rhizomes and stolons become large and difficult to kill.

Use the high rate under heavy grass pressure and/or when grasses are at or near maximum height.

Always add a crop oil concentrate at 1 qt./A by ground or 1% v/v (but not less than 1 pt./A) to the finished spray volume by air.

INSTRUCTIONS FOR ANNUAL BLUEGRASS CONTROL WITH CLETHODIM 2E

GRASS SPECIES	WEED STAGE	RATE FL. OZ./ACRE (lb a.i./A)	HIGH RATE FL. OZ./ACRE (lb a.i./A)
Annual Bluegrass (<i>Poa annua</i>)	To 4-leaf	6* [0.09375]	16 [0.25]

Apply under favorable soil moisture and humidity, which exists within a few days after rainfall or within 7 days after irrigation. Grass needs to be actively growing at time of application(s).

Apply at weed stage indicated on the label, as reduced control can be expected with more mature annual bluegrass.

Use the high rate under heavy grass pressure and/or when annual bluegrass is more mature.

Always add a crop oil concentrate at 1 qt./A by ground to the finished spray volume.

*Use a minimum of 10 fl. oz./A [0.156 lb a.i./A] to control annual bluegrass in seedling and established alfalfa and peppermint and spearmint tops.

DIRECTIONS FOR REDUCED RATE TO CONTROL SMALL ANNUAL GRASSES IN CANOLA, DRIED SHELLED BEAN & PEA (INCLUDING SOYBEAN), EDIBLE PODDED LEGUME VEGETABLES, FLAX, MUSTARD SEED, BEAN & PEA (SUCCULENT SHELLED) AND SUGAR BEET (REDUCED RATE DIRECTIONS NOT FOR USE IN CALIFORNIA)

- Apply only to actively growing grasses at specified weed heights.
- Apply when the first grass weed species in a mixed grass weed population reaches the specified growth stage for treatment.
- Regrowth by tillering may occur if application is made when plants are stressed by lack of moisture, excessive moisture, low or high temperatures and/or under very low humidity.

GRASS SPECIES	SCIENTIFIC NAME	WEED HEIGHT (inches)	RATE FL. OZ./ACRE* (lb a.i./A)
Barnyardgrass	<i>Echinochloa crus-galli</i>	1 to 4	4 [0.0625]
Broadleaf Signalgrass	<i>Brachiaria platyphylla</i>	1 to 4	5 [0.078]
Crabgrass			
Large	<i>Digitaria sanguinalis</i>	1 to 3*	4 [0.0625]
Large	<i>Digitaria sanguinalis</i>	1 to 4*	5 [0.078]
Smooth	<i>Digitaria ischaemum</i>	1 to 3*	4 [0.0625]
Smooth	<i>Digitaria ischaemum</i>	1 to 4*	5 [0.078]
Southern	<i>Digitaria ciliaris</i>	1 to 4*	5 [0.078]
Fall Panicum	<i>Panicum dichotomiflorum</i>	1 to 4	4 [0.0625]
Foxtail			
Giant	<i>Setaria faberi</i>	1 to 4	4 [0.0625]
Green	<i>Setaria viridis</i>	1 to 4	4 [0.0625]
Millet	<i>Setaria italica</i>	1 to 4	5 [0.078]
Yellow	<i>Setaria glauca</i>	1 to 4	4 [0.0625]
Seedling Johnsongrass	<i>Sorghum halepense</i>	1 to 6	5 [0.078]
Shattercane	<i>Sorghum bicolor</i>	4 to 10	4 [0.0625]
Texas Panicum	<i>Panicum texanum</i>	1 to 4	5 [0.078]
Volunteer Cereals			
Barley	<i>Hordeum vulgare</i>	1 to 4	5 [0.078]
Oats	<i>Avena sativa</i>	1 to 4	5 [0.078]
Wheat	<i>Triticum aestivum</i>	1 to 4	5 [0.078]
Volunteer Corn**	<i>Zea mays</i>	4 to 12	4 [0.0625]
Wild Proso Millet	<i>Panicum miliaceum</i>	1 to 6	4 [0.0625]
Wild Oats	<i>Avena fatua</i>	1 to 4	5 [0.078]

*Length of lateral growth

**Not S.R. Corn

† Always add a crop oil concentrate at 1 qt./A by ground application to the finished spray volume.

INSTRUCTIONS FOR PERENNIAL GRASSES

- Apply only to actively growing grasses at specified weed heights.
- Apply when the first grass weed species in a mixed grass weed population reaches the specified growth stage for treatment.
- Use the high rate under heavy grass pressure and/or when grasses are at maximum height.

Restrictions

- **DO NOT** exceed the maximum rate per application listed in the table Crop Specific Use Directions and Restrictions for **Clethodim 2E**.
- **DO NOT** exceed the maximum number of applications per year listed in the table Crop Specific Use Directions and Restrictions for **Clethodim 2E**.
- **DO NOT** exceed the maximum yearly rate listed in the table Crop Specific Use Directions and Restrictions for **Clethodim 2E**.

GRASS SPECIES	WEED HEIGHT (inches)	RATE FL. OZ./ACRE (lb a.i./A)	HIGH RATE FL. OZ./ACRE (lb a.i./A)
Bermudagrass (<i>Cynodon dactylon</i>)			
First Application	3 (or up to 6" runners)	8 [0.125]	16 [0.25]
Repeat Application(s) (if regrowth occurs)	3 (or up to 6" runners)	8 [0.125]	16 [0.25]
Fescue, Tall (<i>Festuca arundinacea</i>)			
First Application	4 to 8	8 [0.125]	16 [0.25]
Repeat Application(s) (if regrowth occurs)	4 to 8	8 [0.125]	16 [0.25]
Foxtail Barley (<i>Hordeum jubatum</i>)			
First Application	2 to 6	8 [0.125]	16 [0.25]
Repeat Application(s) (if regrowth occurs)	2 to 6	8 [0.125]	16 [0.25]
Orchardgrass (<i>Dactylis glomerata</i>)			
First Application	4 to 8	8 [0.125]	16 [0.25]
Repeat Application(s) (if regrowth occurs)	4 to 8	8 [0.125]	16 [0.25]
Quackgrass (<i>Elytrigia repens</i>)			
First Application	4 to 12	8 [0.125]	16 [0.25]
Repeat Application(s) (if regrowth occurs)	4 to 12	8 [0.125]	16 [0.25]
Rhizome Johnsongrass (<i>Sorghum halepense</i>)			
First Application	12 to 24	8 [0.125]	16 [0.25]
Repeat Application(s) (if regrowth occurs)	6 to 18	8 [0.125]	8 [0.125]
Perennial Bluegrass*			
Roughstalk (<i>Poa trivialis</i>)			
Kentucky (<i>Poa pratensis</i>)			
First Application	2 to 4	8 [0.125]	16 [0.25]
Repeat Application(s) (if regrowth occurs)	2 to 4	8 [0.125]	16 [0.25]
Bentgrass* (<i>Agrostis</i> spp.)			
First Application	2 to 4	--	16 [0.25]
Repeat Application(s) (if regrowth occurs)	2 to 4	--	16 [0.25]

*Control of quackgrass, perennial bluegrass and bentgrass with **Clethodim 2E** may be enhanced by adding AMS at 2.5 to 4.0 lbs./A.

TANK MIXES INFORMATION

It is the pesticide user's responsibility to ensure that all products are registered for the intended use. Read and follow the applicable restrictions and limitations and directions for use on all product labels involved in tank mixing. Users must follow the most restrictive directions for use and precautionary statements of each product in the tank mixture, including all crop rotational and other crop restrictions. Those concerns may include, but are not limited to:

1. Geographic restrictions - all products are not registered for use in all areas and rates may vary from one region of labeled use to another;
2. Crop rotation restrictions;
3. Applicator certification requirements;
4. Worker safety rules (e.g. protective clothing, reentry time, posting);

5. Soil type or soil characteristics (e.g. pH, OM);
6. Maximum dosage or number of applications per year;
7. Rain free period required; or
8. Application timing (e.g. pre-harvest interval)
9. **DO NOT** exceed the total yearly rates.

TANK MIX APPLICATION OF CLETHODIM 2E AND BROADLEAF HERBICIDES FOR CONTROL OF GRASSES AND BROADLEAF WEEDS

- Apply only to actively growing grass and broadleaf weeds at specified height or growth stage listed on each label.
- Apply when the first grass or broadleaf weed species in a mixed population reaches the specified height or growth stage for treatment.
- Apply under favorable soil moisture and humidity that exist a few days after rainfall or within seven days after irrigation.
- Always add the appropriate adjuvant to the spray mix at the rate specified for each specific tank mix combination.
- Tank mix applications may sometimes result in reduced grass control and possible increases in crop injury as compared to either product used alone. If regrowth occurs, or an additional flush of new grass emerges, make a second application of **Clethodim 2E**, as specified in the respective size and rate tables.
- **DO NOT** tank mix **Clethodim 2E** when broad leaf weeds are tall and/or dense enough to prevent proper grass coverage.

MIXING INSTRUCTIONS

1. Fill clean spray tank 1/2 to 2/3 of desired level with clean water.
2. While agitating, add the correct amount of **Clethodim 2E**. Agitation must create a rippling or rolling action on the water surface.
3. If tank mixing **Clethodim 2E** with other labeled herbicides, add water soluble bags first, followed by dry formulations, flowables, emulsifiable concentrates, and then solutions. Prepare no more spray mixture than is required for the immediate spray operation.
4. Add any required adjuvants (crop oil concentrate, non-ionic surfactant and/or nitrogen solution).
5. Fill spray tank to desired level with water. Agitation must continue until all spray solution has been applied.

Failure to agitate the spray solution may result in improper mixing of the herbicides and unsatisfactory weed control. Mixing and compatibility qualities must be verified by a jar test.

INFORMATION ON ANTAGONISM

Tank mixes of **Clethodim 2E** with postemergence broadleaf herbicides have shown some reduction or failure to control certain grass species which would have otherwise been controlled when **Clethodim 2E** is applied alone. Activity of the postemergence broadleaf herbicide in the tank mix is not affected.

ALFALFA

Table 1. CLETHODIM 2E TANK MIXES WITH BROADLEAF HERBICIDES FOR ALFALFA
(Refer to the instruction tables above for specific grasses and growth stages)

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + 2,4-DB ⁴	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1%	1%
Clethodim 2E + Imazethapyr ⁵ or Ammonium Salt of Imazethapyr ⁵	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label or See tank mix partner label	--	1%	1%

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Bromoxynil octanoate ⁶ or Bromoxynil heptanoate + Bromoxynil octanoate ^{6,7}	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label or See tank mix partner label	--	0.5%	0.5%

¹ If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (Without a tank mix herbicide), according to the appropriate size and rate directions.

² Broadleaf weed control may be reduced when grass populations are tall or dense enough to intercept the spray pattern and prevent them from receiving complete coverage. Tank mixing is not advised in these situations.

³ Always use a crop oil concentrate at the listed rate (but not less than 1 pt./A) in the finished spray volume.

⁴ **Clethodim 2E** plus 2,4-DB may increase the severity of crop injury when tank mixed. Alfalfa plants will generally outgrow this temporary crop injury within a few weeks.

⁵ Before using this tank mix, read and understand the tank mix partner label for geographical restrictions and restrictions regarding alfalfa growth stage and type. Failure to do so can result in crop injury to alfalfa. **DO NOT** feed, graze, or harvest alfalfa for 30 days following an application of Ammonium salt of imazethapyr to alfalfa.

⁶ In the states of Washington, Oregon, Idaho, Montana, Wyoming, Colorado, Utah, Nevada and the western halves of North Dakota, South Dakota, Nebraska, and Kansas: The **Clethodim 2E** plus Bromoxynil octanoate or Bromoxynil heptanoate + Bromoxynil octanoate tank mix must be applied in the fall or spring to seedling alfalfa when the majority of the field has a minimum of 2 trifoliates. Unacceptable crop injury may occur to alfalfa seedlings less than the 2 trifoliolate leaf stage. **Clethodim 2E** plus Bromoxynil octanoate or Bromoxynil heptanoate + Bromoxynil octanoate applications made when temperatures are expected to exceed 80°F at (and 3 days following) application can result in unacceptable crop injury. In the states not listed above, apply in the fall or spring to seedling alfalfa when the majority of the field has a minimum of 4 trifoliolate leaves. When alfalfa stand is uneven and conditions favor leaf burn, unacceptable crop injury may occur to alfalfa in the 2 trifoliolate or smaller stage at growth. **Clethodim 2E** plus Bromoxynil octanoate or Bromoxynil heptanoate + Bromoxynil octanoate applications made when temperatures are expected to exceed 70°F at (and 3 days following) application can result in unacceptable crop injury. Crop leaf burn can occur following **Clethodim 2E** plus Bromoxynil octanoate or Bromoxynil heptanoate + Bromoxynil octanoate application. Warm, humid conditions may enhance leaf burn. New crop growth will not be affected.

⁷ **DO NOT** apply when alfalfa is under moisture, temperature, insect or disease stress or has been stressed by other pesticide carryover or application.

Table 2. REDUCED RATE CLETHODIM 2E TANK MIXES WITH BROADLEAF HERBICIDES FOR RAPESEED SUBGROUP 20A LISTED CROPS
(Refer to the reduction tables above for specific grasses and growth stages.)

PRODUCT	APPLICATION RATES/ACRE		AMMONIUM SULFATE	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E² + Glufosinate ³	4 to 5 fl. oz. (0.0625 to 0.078 lb a.i.) + See tank mix partner label	--	3.0 lbs.	3.0 lbs.

¹ Annual grasses and sizes controlled with these tank mixtures are those that are identified in the **DIRECTIONS FOR REDUCED RATE TO CONTROL SMALL ANNUAL GRASSES** table.

² **DO NOT** apply **Clethodim 2E** tank mix during or after bolting or flowering or crop injury may occur.

³ For use only on LibertyLink® rapeseed subgroup 20A listed crops

COTTON

Table 3. CLETHODIM 2E TANK MIXED WITH LACTOFEN AND MSMA APPLIED POST DIRECTED TO COTTON

PRODUCT ¹	APPLICATION RATES/ACRE ²		CROP OIL CONCENTRATE ³ V/V	COMMENTS
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	
Clethodim 2E ⁴ + Lactofen + MSMA (4.0 lbs./gal.) or MSMA (6.0 lbs./gal.)	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.)	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.)	1%	Reduce broadcast rate in proportion to the band area actually treated.
	See tank mix partner label for rates to control broadleaf weeds and height limitations for cotton. Refer to the Clethodim 2E label for weed height and species control.			
	See tank mix partner label for rates to control broadleaf weeds and height limitations for cotton. Refer to the Clethodim 2E label for weed height and species control.			

¹ Broadleaf weed control may be reduced when grass populations are tall or dense enough to intercept the spray pattern and prevent them from receiving complete coverage. Tank mixing is not advised in these situations.

² If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (without a tank mix herbicide), according to the appropriate size and rate directions.

³ Always use a crop oil concentrate at the listed rate (but not less than 1 pt./A) in the finished spray volume.

⁴ If at the time of application, grass height is so tall that post-directed applications cannot get good coverage over the top of the grassy weeds, then poor control may result and a second (non-post directed) application of **Clethodim 2E** may be necessary.

Table 4. CLETHODIM 2E TANK MIXED WITH BROMOXNYL HEPTANOATE + BROMOXNYL OCTANOATE TO CONTROL EMERGED WEEDS IN BXN COTTON AS A BROADCAST APPLICATION

PRODUCT	APPLICATION RATES/ACRE	CROP OIL CONCENTRATE PER ACRE ³	COMMENTS ⁷
	ANNUAL GRASSES		
Clethodim 2E² + Bromoxynil heptanoate + Bromoxynil octanoate ^{4,5,6}	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.)	1 qt.	See charts for grasses controlled.
	See tank mix partner label for rates to control broadleaf weeds and height limitations for cotton		

¹ Broadleaf weed control may be reduced with grass populations are tall or dense enough to intercept the spray pattern and prevent them from receiving complete coverage.

² If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** at the specified rate with the appropriate amount of crop oil concentrate in a non-Bromoxynil heptanoate + Bromoxynil octanoate tank mix.

³ Always add a crop oil concentrate at 1 qt./A by ground in the finished spray solution.

⁴ Applications of Bromoxynil heptanoate + Bromoxynil octanoate can be made only to cotton that has been genetically modified for crop resistance to postemergence over-the-top applications of bromoxynil.

⁵ **RESTRICTION: DO NOT** apply the **Clethodim 2E** plus Bromoxynil heptanoate + Bromoxynil octanoate tank mix within 75 days of harvest.

⁶ **RESTRICTION: DO NOT** exceed 2 applications of Bromoxynil heptanoate + Bromoxynil octanoate before cotton is 12 inches tall and one application after 12 inches tall.

⁷ Use a minimum of 10 gals. of spray solution per acre.

Table 5. CLETHODIM 2E TANK MIXED WITH GLYPHOSATE TO CONTROL EMERGED GRASSES IN COTTON AS A BROADCAST APPLICATION

PRODUCT	APPLICATION RATES/ACRE		ADJUVANT		COMMENTS
	ANNUAL GRASSES	PERENNIAL GRASSES	Glyphosate formulation with built in adjuvant	Glyphosate formulation without built in adjuvant	
Clethodim 2E + Glyphosate	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.)	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.)	Non-ionic surfactant @ 0.125 to 0.25% v/v plus ammonium sulfate @ 8.5 to 17 lbs. per 100 gals. of carrier	Non-ionic surfactant @ 1 pt./A plus ammonium sulfate @ 8.5 to 17 lbs. per 100 gals. of carrier	See charts for grasses controlled. Use a minimum of 10 gals. of spray solution per acre.

¹ If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** at the specified rate with the appropriate amount of crop oil concentrate

DRY SHELLED AND SUCCULENT BEANS

Table 6. CLETHODIM 2E TANK MIXES WITH BROADLEAF HERBICIDES FOR DRY SHELLED AND SUCCULENT BEANS
(Refer to the recommendation tables above for specific grasses and growth stages.)

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Bentazon	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1%	1%

¹ If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (without a tank mix herbicide), according to the appropriate size and rate directions.

² Broadleaf weed control may be reduced when grass populations are tall enough or dense enough to intercept the spray pattern and prevent them from receiving complete coverage. Tank mixing is not advised in these situations.

³ Always use a crop oil concentrate at the listed rate (but not less than 1 pt./A) in the finished spray volume.

FLAX

Table 7. REDUCED RATE CLETHODIM 2E TANK MIXES WITH BROADLEAF HERBICIDES FOR FLAX
(Refer to the direction tables above for specific grasses and growth stages)

PRODUCT	APPLICATION RATES/ACRE		CROP OIL CONCENTRATE	
	ANNUAL GRASSES ¹	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Bromoxynil heptanoate + Bromoxynil octanoate + 2-ethylhexyl ester of MCPA ^{2,3}	4 to 5 fl. oz. (0.0625 to 0.078 lb a.i.) + See tank mix partner label	--	1 pt.	1 pt.

PRODUCT	APPLICATION RATES/ACRE		CROP OIL CONCENTRATE	
	ANNUAL GRASSES ¹	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Bromoxynil octanoate	4 to 5 fl. oz. (0.0625 to 0.078 lb a.i.) + See tank mix partner label	--	1 pt.	1 pt.
Clethodim 2E + 2-ethylhexyl ester of 2-methyl-4-chlorophenoxyacetic acid ^{2,3}	4 to 5 fl. oz. (0.0625 to 0.078 lb a.i.) + See tank mix partner label	--	1 pt.	1 pt.

¹ Annual grasses and sizes controlled with these tank mixtures are those that are identified in the **DIRECTIONS FOR REDUCED RATE TO CONTROL SMALL ANNUAL GRASSES** table.

² **DO NOT** apply **Clethodim 2E** tank mix during or after the bud stage or to ornamental flax or crop injury may occur.

³ **DO NOT** apply tank mixes if temperatures are expected to exceed 85°F at (or 3 days following) application or crop injury may occur.

SOYBEAN

Table 8. CLETHODIM 2E TANK MIXES² TO CONTROL ANNUAL GRASSES WHEN USED AS A BURNDOWN IN NO-TILL SOYBEAN

PRODUCT	PRODUCT RATE/ ACRE ¹	GRASS HEIGHT (Inches)	CROP OIL CONCENTRATE/ACRE ²	28%N OR 32%N QTS./A OR 2.5 TO 4.0 LBS. AMS
Clethodim 2E + 2,4-D Ester* ³	3 fl. oz. (0.0468 lb a.i.)	Foxtail 1 to 3 Fall Panicum 1 to 3	1 qt.	1 to 2 qts. or 2.5 to 4.0 lbs. AMS
	4 fl. oz. (0.0625 lb a.i.)	Foxtail 1 to 4 Fall Panicum 1 to 4	1 qt.	1 to 2 qts. or 2.5 to 4.0 lbs. AMS
	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	(See Grass Chart for grasses claimed.)	1 qt.	1 to 2 qts. or 2.5 to 4.0 lbs. AMS

*2,4-D ester must not be used where drift sensitive crops may be grown.

¹ If regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** according to the appropriate size and rate directions.

² Always use a crop oil concentrate at the listed rate in the finished spray volume.

³ The following products can be tank mixed with **Clethodim 2E** plus 2,4-D ester: Flumioxazin, Sulfentrazone + Chlorimuron Ethyl, Metolachlor, S-metolachlor, Pendimethalin, Metribuzin.

Table 9. CLETHODIM 2E TANK MIXES WITH BROADLEAF HERBICIDES FOR SOYBEAN
(Refer to the direction tables above for specific grasses and growth stages.)

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CON- CENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Lactofen	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	0.5 to 1%	1%
Clethodim 2E + Bentazon	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1%	1%
Clethodim 2E + Glyphosate (for use on Roundup Ready soybean only)	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	0.5 to 1% ⁴	1% ⁴
Clethodim 2E + Acifluorfen	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	0.5 to 1%	1%
Clethodim 2E + Fomesafen	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	1%	1%
Clethodim 2E + Chlorimuron ethyl	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1%	1%
Clethodim 2E + Imazethapyr	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	1%	1%
Clethodim 2E + Lactofen + Chlorimuron ethyl	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label + See tank mix partner label	--	0.5%	1%
Clethodim 2E + Lactofen + Bentazon	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label + See tank mix partner label	--	0.5%	1%

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CON- CENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Lactofen + Imazethapyr	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label + See tank mix partner label	--	0.5%	1%
Clethodim 2E + Bentazon + Acifluorfen	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label	--	0.5%	1%
Clethodim 2E + Flumiclorac pentyl ester + Imazethapyr	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + 4 fl. oz. + See tank mix partner label	--	1%	1%
Clethodim 2E + Flumiclorac pentyl ester + Bentazon	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label + See tank mix partner label	--	1%	1%
Clethodim 2E + Flumiclorac pentyl ester + Chlorimuron-ethyl	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label + See tank mix partner label	--	1%	1%
Clethodim 2E + Lactofen + Flumiclorac pentyl ester	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label + See tank mix partner label	--	0.5%	1%
Clethodim 2E + Cloransulam-methyl	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	1%	--
Clethodim 2E + Lactofen + Cloransulam-methyl	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label + See tank mix partner label	1%	--

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CON- CENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Imazamox	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	--	1%	--
Clethodim 2E + Lactofen + Imazamox	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label + See tank mix partner label	--	1%	--
Clethodim 2E + Chlorimuron ethyl + Thifensulfuron methyl	6 to 8 fl. oz. ⁷ (0.09375 - 0.125 lb a.i.) + See tank mix partner label	--	1 qt.	--
Clethodim 2E + Lactofen + Chlorimuron ethyl + Thifenusulfuron methyl	6 to 8 fl. oz. ⁷ (0.09375 - 0.125 lb a.i.) + See tank mix partner label + See tank mix partner label	--	1 pt.	--
Clethodim 2E + Flumiclorac pentyl ester	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	--	1 qt.	--
Clethodim 2E + Chloransulam-methyl + Flumetsulam	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label	--	1%	--
Clethodim 2E + Cloransulam-methyl + Fomesafen	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label + See tank mix partner label	1%	--

¹ If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (without a tank mix herbicide), according to the appropriate size and rate directions.

² Broadleaf weed control may be reduced when grass populations are tall or dense enough to intercept the spray pattern and prevent them from receiving complete coverage. Tank mixing is not advised in these situations.

³ Always use a crop oil concentrate at the listed rate (but not less than 1 pt./A) in the finished spray volume.

⁴ The addition of 2.5 lb. of ammonium sulfate is required when **Clethodim 2E** is tank mixed with glyphosate. If the glyphosate formulation has a standalone built in adjuvant, add 0.125% v/v non-ionic surfactant in place of crop oil concentrate. If the glyphosate formulation does not have a built in adjuvant system, add 0.5 to 1% crop oil concentrate for ground application and 1 % v/v for aerial application.

⁵ The addition of 1 to 2 qts./A of liquid fertilizer (10-34-0, 28%N or 32%N) is advised when **Clethodim 2E** is tank mixed with Imazethapyr, Flumiclorac pentyl ester, Bentazon + Acifluorfen, Cloransulam-methyl, Chlorimuron ethyl + Thifensulfuron methyl, Ammonium salt of imazamox, Chloransulam-methyl + Flumetsulam, Lactofen plus Chlorimuron ethyl, Lactofen plus Bentazon, Lactofen plus Imazethapyr, Lactofen plus Cloransulam-methyl, Lactofen plus Chlorimuron ethyl + Thifendulfuron methyl, and Lactofen plus Imazamox. An equivalent amount (2.5 to 4.0 lbs./A) of spray grade ammonium sulfate (AMS) may be added in place of liquid fertilizer. Fertilizer adjuvants are to be added in addition to the crop oil concentrate.

⁶ Refer to tank mix partner label for geographic and rotational restrictions.

⁷ Annual grasses and sizes controlled with these tank mixtures are those that are identified in the **DIRECTIONS FOR REDUCED RATE TO CONTROL SMALL ANNUAL GRASSES** table.

Table 10. REDUCED RATE CLETHODIM 2E TANK MIXES WITH BROADLEAF HERBICIDES FOR SOYBEAN

(Refer to table for reduced rate use in rapeseed subgroup 20A listed crops, dry shelled bean & pea, edible podded legume vegetables, flax, mustard seed, soybean, succulent shelled bean & pea and sugar beet directions for small annual grasses for specific grasses and growth stages)

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Cloransulam-methyl	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	--	1%	1%
Clethodim 2E + Ammonium salt of imazethapyr	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	--	1%	1%

¹ If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (without a tank mix herbicide), according to the appropriate size and rate directions.

² Annual grasses and sizes controlled with these tank mixtures are those that are identified in the **DIRECTIONS FOR REDUCED RATE TO CONTROL SMALL ANNUAL GRASSES** table.

³ Always use a crop oil concentrate at the listed rate (but not less than 1 pt./A) in the finished spray volume.

⁴ The addition of 1 to 2 qts./A of liquid fertilizer (10-34-0, 28%N, or 32%N) is required when **Clethodim 2E** is tank mixed at reduced rates. An equivalent amount (2.5 to 4.0 lbs./A) of spray grade ammonium sulfate (AMS) may be added in place of liquid fertilizer. Fertilizer adjuvants are to be added in addition to the crop oil concentrate.

PEANUT (INCLUDING PERENNIAL)

Table 11. CLETHODIM 2E TANK MIXES WITH BROADLEAF HERBICIDES FOR PEANUT (INCLUDING PERENNIAL)

(Refer to the direction tables above for specific grasses and growth stages.)

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Bentazon	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label	--	1%	1%
Clethodim 2E + Acifluorfen	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label	--	1%	1%
Clethodim 2E + Bentazon + Acifluorfen	8 to 10 fl. oz. (0.125 to 0.156 lb a.i.) + See tank mix partner label	--	1%	1%

¹ If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (without a tank mix herbicide), according to the appropriate size and rate directions.

² Broadleaf weed control may be reduced when grass populations are tall or dense enough to intercept the spray pattern and prevent them from receiving complete coverage. Tank mixing is not advised in these situations.

³ Always use a crop oil concentrate at the listed rate (but not less than 1 qt./A) in the finished spray volume.

DIRECTIONS FOR GRASS SUPPRESSION FOR HARVEST EFFICIENCY IN PEANUT (INCLUDING PERENNIAL) WITH CLETHODIM 2E				
GRASS SPECIES		WEED STAGE	RATE FL. OZ./A (LB A.I./A)	HIGH RATE (LB A.I./A)
Annual and perennial grasses that exceed height claimed for control on height charts " DIRECTIONS FOR ANNUAL GRASSES " & " DIRECTIONS FOR PERENNIAL GRASSES "		Up to and including grasses in the seed head stage	16 (0.25)	32 (0.5)
DO NOT apply as part of a tank mix when applying Clethodim 2E for grass suppression. Add a crop oil concentrate at 1 qt./A by ground to the finished spray volume.				

SUGAR BEET

Table 12. CLETHODIM 2E TANK MIXED WITH CLOPYRALID APPLIED TO SUGAR BEET

(Refer to the direction tables above for specific grasses and growth stages)

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE ³ V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + Clopyralid	6 to 8 fl. oz./A (0.09375 - 0.125 lb a.i./A)	8 to 16 fl. oz./A (0.125 to 0.25 lb a.i./A)	1%	1%
See tank mix partner label for rates.				

¹ If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (without a tank mix herbicide), according to the appropriate size and rate directions.

² Broadleaf weed control may be reduced when grass populations are tall or dense enough to intercept the spray pattern and prevent them from receiving complete coverage. Tank mixing is not advised in these situations.

³ Always use a crop oil concentrate at the listed rate (but not less than 1 pt./A) in the finished spray volume.

Table 13. CLETHODIM 2E TANK MIXED WITH BETAMIX® or BETANEX® APPLIED TO SUGAR BEET

PRODUCT ¹	WEEDS CONTROLLED		WEED HEIGHT (inches)	APPLICATION RATE/ACRE ²
	COMMON NAME	SCIENTIFIC NAME		
Clethodim 2E³ + Phenmedipham + Desmedipham or Desmedipham	Barnyardgrass	<i>Echinochloa crus-galli</i>	1 to 3	8 fl. oz. (0.125 lb a.i.)
	Foxtail	<i>Setaria</i> spp.	1 to 3	
	Foxtail Millet	<i>Setaria italic</i>	1 to 3	
	Wild Oat	<i>Avena fatua</i>	1 to 3	See tank mix partner label for rates to control broadleaf weeds. No additives are advised in the tank mix.
	Wild Proso Millet	<i>Panicum miliaceum</i>	1 to 3	See tank mix partner label for rates to control broadleaf weeds. No additives are advised in the tank mix.

¹ Broadleaf weed control may be reduced when grass populations are tall or dense enough to intercept the spray pattern and prevent them from receiving complete coverage. Tank mixing is not advised in these situations.

² **DO NOT** use crop oil concentrate. No additives are advised in the tank mix. If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (without a tank mix herbicide), according to the appropriate size and rate directions.

³ If grass regrowth occurs or an additional flush of new grass emerges, make a second application of **Clethodim 2E** at full label rate with appropriate rate of crop oil concentrate.

Table 14. CLETHODIM 2E PLUS BETANEX OR BETAMIX TANK MIX FOR THREE SEQUENTIAL APPLICATIONS FOR ANNUAL GRASS CONTROL (MICRO-RATE APPLICATION)

PRODUCT ²	APPLICATION RATES/ACRE ¹		METHYLATED SEED OIL ² (V/V)	
	ANNUAL GRASSES	GRASSES CONTROLLED (inches)	GROUND	AIR
Clethodim 2E + Phenmedipham + Desmedipham or Desmedipham	2 to 3 fl. oz. (0.031 to 0.0468 lb a.i.) + See tank mix partner label ³ or See tank mix partner label ³	Green Foxtail (1-2) Yellow Foxtail (1-2) Barnyardgrass (1-2) Wild Oat (1-2) Volunteer Cereals (1-2)	1.5%	1.5%

¹ Broadleaf weed control may be reduced when grass populations are tall or dense enough to intercept the spray pattern and prevent them from receiving complete coverage. Tank mixing is not advised in these situations.

² Always use a methylated seed oil at the listed rate (but not less than 1 pt./A) in the finished spray volume.

³ Use 8 fl. oz./A [0.125 lb a.i./A] rate when sugar beet are in the cotyledon to 4 leaf stage. Rate can be increased up to 12 fl. oz./A [0.1875 lb a.i./A] when the smallest sugar beet plants in the field are in the 4 true leaf stage or larger.

Directions for Use for Micro-Rate Applications to Sugar Beet

Product Information

Multiple micro-rate applications of **Clethodim 2E** in tank mixtures with reduced rates of Desmedipham or Phenmedipham + Desmedipham and methylated seed oils may be applied by air or ground equipment to sugar beet to control early germinating annual grasses listed above. Refer to the desmedipham or phenmedipham + desmedipham product labels for use rates. Favorable climatic conditions [good conditions for plant growth and development] are essential for adequate weed control. All use precautions and restrictions on the tank mix partner master labels must be followed.

Directions for Using Micro-Rate Multiple Applications of Clethodim 2E Tank Mixes

Apply **Clethodim 2E** in broadcast applications only at a rate of 2 to 3 fl. oz./A [0.031 to 0.0468 lb a.i./A] in tank mixture with either Desmedipham or Phenmedipham + Desmedipham following the directions for use on the tank mix partner label. A minimum of three sequential applications of 2 fl. oz./A [0.031 lb a.i./A] or a minimum of 2 sequential applications of 3 fl. oz./A [0.0468 lb a.i./A] must be utilized for **Clethodim 2E** tank mixtures. Refer to the desmedipham or phenmedipham + desmedipham product labels for application restrictions. Accurate timing is essential; make initial application immediately after weeds emerge, and make repeat applications on 5 to 7 day intervals. If weed control is not adequate due to climatic conditions, spray coverage or other factors, return to conventional application rates of **Clethodim 2E** [6 to 8 fl. oz./A] [0.09375 - 0.125 lb a.i./A] and add rates of Desmedipham or Phenmedipham + Desmedipham as directed on their label. When using conventional rates of Desmedipham or Phenmedipham + Desmedipham in tank mixtures with **Clethodim 2E**, a spray adjuvant is not advised.

Use Precautions for Micro-Rate Applications: [See Clethodim 2E, Desmedipham or Phenmedipham + Desmedipham master label for further use precautions.]

Not all weeds will be adequately controlled, even with favorable climatic conditions. Conventional rates of **Clethodim 2E**, Desmedipham or Phenmedipham + Desmedipham and/or hand labor may be required if multiple micro-rate applications do not adequately control weeds. Plugging of spray nozzles may be encountered due to the potential for formation of a precipitate in the spray solution that is often associated with micro-rate applications. To the extent consistent with applicable law, Tenkoz, Inc. will not be responsible for any nozzle plugging that may occur with the use of multiple micro-rate applications.

GROUND APPLICATION

Use of sufficient spray volumes and pressure is essential to ensure complete coverage. Use a minimum of 10 gals. and a maximum of 20 gals. of spray solution per acre. Spray pressures must reflect a minimum of 30 psi and a maximum of 60 psi at the nozzle. **DO NOT** use flood nozzles.

AERIAL APPLICATION

Use of sufficient spray volumes is essential to ensure complete coverage. Use a minimum of 5 gals. and a maximum of 15 gals. of spray solution per acre.

Table 15. TANK MIX APPLICATION OF CLETHODIM 2E AND FUNGICIDES FOR CONTROL OF GRASS WEEDS AND DISEASES IN SUGAR BEET

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE ³ V/V
	ANNUAL GRASSES	PERENNIAL GRASSES	
Clethodim 2E	6 to 8 fl. oz./A [0.09375 - 0.125 lb a.i./A]	8 to 16 fl. oz./A [0.125 to 0.25 lb a.i./A]	1%
+ Tetraconazole	+ See tank mix partner label	+ See tank mix partner label	

¹ If grass regrowth occurs, or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone [without a tank mix fungicide] according to the appropriate size and rate directions.

² Refer to **Clethodim 2E** and fungicide label for rates and weeds and diseases controlled.

³ Always use a crop oil concentrate at the listed rate [but not less than 1 pt./A] in the finished spray volume.

Table 16. TANK MIX APPLICATION OF CLETHODIM 2E AND INSECTICIDES FOR CONTROL OF GRASS WEEDS AND INSECTS IN ALFALFA, COTTON, PEPPERMINT AND SPEARMINT TOPS, PEANUT (INCLUDING PERENNIAL), SOYBEAN AND SUNFLOWER

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE (V/V) ³	CROP					
	ANNUAL GRASSES	PERENNIAL GRASSES		Alfalfa ⁴	Cotton	Peppermint & Spearmint Tops ^{4, 5}	Peanut	Soybean	Sunflower
Clethodim 2E + Acephate ⁴	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	1%		X	X	X	X	
Clethodim 2E + Fenpropathrin	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	1%		X		X		
Clethodim 2E + Esfenvalerate	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	1%						X
Clethodim 2E + lamda-Cyhalothrin	6 to 8 fl. oz. (0.09375 - 0.125 lb a.i.) + See tank mix partner label	8 to 16 fl. oz. (0.125 to 0.25 lb a.i.) + See tank mix partner label	1%						X

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE (V/V) ³	CROP					
	ANNUAL GRASSES	PERENNIAL GRASSES		Alfalfa ⁴	Cotton	Peppermint & Spearmint Tops ^{4, 5}	Peanut	Soybean	Sunflower
Clethodim 2E + lamda- Cyhalothrin	10 to 16 fl. oz. ⁷ (0.156 to 0.25 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1%	X					
Clethodim 2E + Beta- cyfluthrin	10 to 16 fl. oz. ⁷ (0.156 to 0.25 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1%	X					
Clethodim 2E + Dimethoate	10 to 16 fl. oz. ⁷ (0.156 to 0.25 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1%	X					
Clethodim 2E + Chlorpyrifos	10 to 16 fl. oz. ⁷ (0.156 to 0.25 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1 to 2 pt. ⁸	X					

PRODUCT ²	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE (V/V) ³	CROP					
	ANNUAL GRASSES	PERENNIAL GRASSES		Alfalfa ⁴	Cotton	Peppermint & Spearmint Tops ^{4,5}	Peanut	Soybean	Sunflower
Clethodim 2E + Permethrin	10 to 16 fl. oz. ⁷ (0.156 to 0.25 lb a.i.) + See tank mix partner label	10 to 16 fl. oz. (0.156 to 0.25 lb a.i.) + See tank mix partner label	1%	X					

¹If grass regrowth occurs, or an additional flush of new grass emerges, make a second application of **Clethodim 2E** alone (without a tank mix insecticide) according to the appropriate size and rate directions.

²Refer to **Clethodim 2E** and insecticide label for rates and insects controlled.

³Always use a crop oil concentrate at the listed rate (but not less than 1 pt./A) in the finished spray volume.

⁴Certain insecticides may cause temporary phytotoxic symptoms on alfalfa and peppermint and spearmint tops foliage. Refer to the insecticide label for further information. It is suggested that prior to using any of these insecticide/herbicide tank mixtures, that a small area of the field be treated first and observations for crop injury be made prior to treating the whole field.

⁵The **Clethodim 2E** rate must be 6 to 8 fl. oz./A (0.09375 - 0.125 lb a.i./A) for annual grass control in baby peppermint and spearmint tops, minimum of 8 fl. oz./A (0.125 lb a.i./A) for annual grass control in established peppermint and spearmint tops and 8 to 16 fl. oz./A for (0.125 to 0.25 lb a.i./A) perennial grass control. Crop oil concentrate must be added at the rate of 1.0 to 2.0 pts./A.

⁶Insecticide tank mix use with Acephate in soybean is permitted only in a state having an approved Section 24(c) registration for Acephate use in soybean.

⁷The **Clethodim 2E** rate must be 6 to 8 fl. oz./A (0.09375 - 0.125 lb a.i./A) for annual grass control in seedling alfalfa.

⁸For the **Clethodim 2E** plus Chlorpyrifos tank mix, reduce the adjuvant rate down to 1.0 pt./A when the Chlorpyrifos rate is 1.0 pt./A or higher.

Table 17. DIRECTIONS FOR ROUNDUP READY VOLUNTEER CORN CONTROL IN ROUNDUP READY SOYBEAN WITH CLETHODIM 2E TANK MIX

Roundup Ready Volunteer Corn Height (inches)	Clethodim 2E Rate fl. oz./A (lb a.i./A)	Glyphosate ¹ rate for formulations with built in adjuvant	Adjuvant
<12	4 (0.0625)	Refer to product label for use rates.	Non-ionic surfactant @ 0.125 to 0.25% v/v plus ammonium sulfate @ 8.5 to 17 lbs. per 100 gals. of carrier.
12 to 18	5 (0.078)		
18 to 24	6 (0.09375)		

Roundup Ready Volunteer Corn Height (inches)	Clethodim 2E Rate fl. oz./A	Glyphosate ¹ rate for formulations <u>without</u> built in adjuvant	Adjuvant
<12	4 (0.0625)	Refer to product label for use rates.	Crop oil concentrate @ 1 pt./A plus ammonium sulfate @ 8.5 to 17 lbs. per 100 gals. of carrier.
12 to 18	5 (0.078)		
18 to 24	6 (0.09375)		

¹Glyphosate formulation must be labeled for use on Roundup Ready soybean.

THE MOST RESTRICTIVE LABELING OF ANY PRODUCT USED IN A TANK MIX MUST BE FOLLOWED.

- Apply only to actively growing grass and broadleaf weeds at specified height or growth stage listed on each label.
- Apply under favorable soil moisture and humidity which exist a few days after rainfall or within seven days after irrigation.

- Tank mix applications may sometimes result in reduced grass control. If regrowth occurs, or an additional flush of new grass emerges, make a second application of **Clethodim 2E**, as specified in the respective size and rate tables.
- This tank mix may be applied postemergence to ROUNDUP READY soybean up through the full flowering stage. Avoid contact with foliage, green stems, or fruit crops, or any desirable plants and trees, other than soybean with the ROUNDUP READY gene as severe injury or destruction will result.

Restrictions

- **DO NOT** tank mix **Clethodim 2E** when broadleaf weeds are tall and/or dense enough to prevent proper grass coverage.
- **DO NOT** apply less than 60 days before harvest.
- **DO NOT** allow the **Clethodim 2E** plus ROUNDUP to mist, drip, drift or splash onto desirable vegetation as minute quantities of the tank mix can cause severe damage or destruction to the crops, plants or other areas on which treatment was not intended. The likelihood of injury occurring from drift of this product is greatest when winds are gusty or in excess of 5 miles per hour. Even under lesser wind velocities, avoid conditions that allow spray drift to occur including combinations of spray pressure and nozzle type that will result in fine particles (mist) that are likely to drift.

CONIFER TREES SPECIFIC RESTRICTIONS AND DIRECTIONS FOR CLETHODIM 2E			
CROPS		Use Rate Per Acre	Special Use Instructions
COMMON NAME	SCIENTIFIC NAME	6-16 fl. oz. (0.09375 - 0.25 lb a.i.)	Clethodim 2E can be used to control labeled grasses in Christmas tree farms, conifer nurseries, and conifer plantations (but not in forests). RESTRICTIONS: • DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. • DO NOT make more than 2 applications per acre per year. • DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Arborvitae, American	<i>Thuja occidentalis</i>		
Cedars	<i>Cedrus</i> spp.		
Cypress	<i>Taxodium</i> spp.		
Fir, Douglas	<i>Pseudotsuga menziesii</i>		
Firs	<i>Abies</i> spp.		
Hemlock, Canadian/Eastern	<i>Tsuga Canadensis</i>		
Hemlock, Western	<i>Tsuga heterophylla</i>		
Pines	<i>Pinus</i> spp.		
Spruces	<i>Picea</i> spp.		
Yew	<i>Taxus</i> spp.		

NON-CROP OR NON-PLANTED AREAS

The following areas are considered non-crop or non-planted areas: rights-of-way including railroads, highways, roads, dividers, medians, pipelines, public utility lines, pumping stations, transformer stations and substations. Around airports, electric utilities, commercial buildings, manufacturing plants, storage yards, rail yards, fence lines, parkways, and post-harvest croplands. Also beneath greenhouse benches and around golf courses.

DIRECTIONS FOR GRASS SUPPRESSION IN NON-CROP AREAS WITH CLETHODIM 2E			
GRASS SPECIES	WEED STAGE	RATE FL. OZ./A (LB A.I./A)	HIGH RATE (LB A.I./A)
Annual and perennial grasses that exceed height claimed for control on height chart above.	Up to and including grasses in the seed head stage	12 (0.1875)	16 (0.25)
DO NOT apply as part of a tank mix when applying Clethodim 2E for grass suppression. Add a crop oil concentrate at 1 qt./A by ground to the finished spray volume.			

Restrictions

- **DO NOT** plant any crop for 30 days after application unless clethodim is registered for use in that crop.
- **DO NOT** apply more than 16 fl oz (0.250 lb ai) per acre per application.
- **DO NOT** make more than 2 applications per acre per year.
- **DO NOT** apply more than 32 fl oz (0.50 lb ai) per acre per year.
- For repeat applications make on a minimum of a 14 day interval.

FALLOW LAND

DIRECTIONS FOR USE

Clethodim 2E may be used to control annual and perennial grasses in land that has been left fallow the previous year and other non-producing agricultural areas. Apply **Clethodim 2E** at 6 to 8 fl. oz./A (0.09375 - 0.125 lb a.i./A) or annual grasses and 8 to 16 fl. oz./A (0.125 to 0.25 lb a.i./A) for perennial grasses. When both grass and broadleaf weeds are the target pest, **Clethodim 2E** may be tank mixed with 2,4-D ester or sodium salt of dicamba for broad spectrum control. When both annual and perennial grasses occur in the same field, use a minimum of 8 fl. oz./A **Clethodim 2E** (0.125 lb a.i./A) rate.

- Use a minimum spray volume of 5 gals./A for aerial applications and 15 gals./A for ground applications.
- Apply only to actively growing grasses when the first grass reaches the specified weed height as specified by the Directions for Annual and Perennial Grasses section of this label.
- Annual grasses that emerge after the **Clethodim 2E** application will not be controlled, and a second application may be necessary.
- The control of perennial grasses may require more than 1 application in non-tilled areas.

Restrictions

- **DO NOT** plant any crop for 30 days after application unless clethodim is registered for use in that crop.
- **DO NOT** apply to grasses that have tillered, formed seedheads or exceeded specified growth stage.
- **DO NOT** use flood jet nozzles.
- **DO NOT** apply to drought stressed grasses.
- **DO NOT** mow area for 2 weeks prior to or after the **Clethodim 2E** application.

Table 18. CLETHODIM 2E IN TANK MIXES TO CONTROL ANNUAL AND PERENNIAL GRASSES IN FALLOW LAND

PRODUCT	APPLICATION RATES/ACRE ¹		CROP OIL CONCENTRATE ² V/V	
	ANNUAL GRASSES	PERENNIAL GRASSES	GROUND	AIR
Clethodim 2E + 2,4-D ester or Sodium salt of dicamba	6 to 8 fl. oz./A (0.09375 - 0.125 lb a.i./A) + See tank mix partner label	8 to 16 fl. oz./A (0.125 to 0.25 lb a.i./A)	1%	1%

¹ Refer to **Clethodim 2E** label for weed height and species controlled. Review Sodium salt of dicamba and 2,4-D labels for crop restrictions, use rates and weeds controlled.

² Always use a crop oil concentrate or methylated seed oil containing at least 15% emulsifier at the listed rate (but not less than 1 pt./A) in the finished spray volume.

TABLE 19. CLETHODIM 2E FOR THE CONTROL AND/OR SUPPRESSION OF TALL FESCUE IN NATIVE PRAIRIE WARM-SEASON GRASS RESTORATION PROJECTS

PRODUCT	PRODUCT RATES	GRASS WEEDS CONTROLLED/SUPPRESSED		WEED STAGES
		Common Name	Scientific Name	
Clethodim 2E	10 to 12 fl. oz./A [0.156 to 0.1875 lb a.i./A]	Tall Fescue	<i>Festuca arundinacea</i>	4 to 6 inches tall [40 to 60% green-up]
Adjuvant: Clethodim 2E must be applied with crop oil concentrate at 1 qt./A, plus a spray grade ammonium sulfate at 2.5 to 4 lbs./A. Advised Mixing order: Thoroughly mix spray grade ammonium sulfate in water, add Clethodim 2E , then add crop oil concentrate.				

SPECIAL APPLICATION INSTRUCTIONS

Burn or mow fields a minimum of 3 weeks prior to application to remove excess crop residue. Apply in the spring, at 40 to 60% tall fescue green-up, prior to emergence of warm-season grasses.

Apply in a minimum of 15 to 20 gals. of water per acre at a spray pressure of 40 to 60 PSI at the nozzle. Apply using flat fan or hollow cone nozzles.

Apply only to fields that have warm-season grasses established for 2 years. Applications of **Clethodim 2E** to emerged warm-season grasses may cause injury.

NOTE: **Clethodim 2E** applications are most effective if applied when average nighttime temperatures are consistently greater than or equal to 47°F.

Restrictions

- **DO NOT** mow area for 2 weeks after the **Clethodim 2E** application.
- **DO NOT** use flood jet nozzles.
- **DO NOT** apply to warm-season grasses grown for seed.
- **DO NOT** graze treated fields or feed treated forage and or hay to livestock.
- **DO NOT** plant any crop for 30 days after application, unless clethodim is registered for use in that crop.

TABLE 20. CLETHODIM 2E FOR THE SUPPRESSION OF TALL FESCUE SEED-HEADS IN NON-PRODUCING AGRICULTURAL AREAS

PRODUCT	PRODUCT RATE	SUPPRESSION	APPLICATION TIMING
Clethodim 2E	1 ½ to 2 fl. oz./A [0.023 to 0.031 lb a.i./A]	Tall Fescue Seed-Heads (<i>Festuca arundinacea</i>)	[50 to 90% Tall Fescue green-up]
ADJUVANT: Clethodim 2E must be applied with crop oil concentrate at 1 qt./A, plus a spray grade ammonium sulfate at 2.5 to 4 lb./A. Advised Mixing order: Thoroughly mix spray grade ammonium sulfate in water, add Clethodim 2E , then add crop oil concentrate.			

SPECIAL APPLICATION INSTRUCTIONS

- Apply at 50 to 90% tall fescue green-up.
- Use the higher **Clethodim 2E** rate if less tall fescue green matter is present.
- Apply in a minimum of 15 to 20 gals. of water per acre at a spray pressure of 40 to 60 psi at the nozzle. Apply using flat fan or hollow cone nozzles. **DO NOT** use flood nozzles.
- 2,4-D ester may be added to this tank mix for broadleaf control [see 2,4-D ester label for weeds controlled].

Restrictions

- **DO NOT** mow area for two weeks after the **Clethodim 2E** application.
- **DO NOT** graze treated fields or feed treated forage and/or hay to livestock.
- **DO NOT** plant any crop for 30 days after application, unless clethodim is registered for use in that crop.

NON-BEARING FRUIT AND NUT CROPS
SPECIFIC RESTRICTIONS AND DIRECTIONS FOR CLETHODIM 2E

CROPS		Use Rate Per Acre	Special Use Instructions
COMMON NAME	SCIENTIFIC NAME	6-8 fl. oz. (0.09375 - 0.125 lb a.i.)	Non-bearing fruit and nut crops are plants which will not bear fruit or nuts for at least one year following Clethodim 2E application. Add a non-ionic surfactant containing at least 80% active ingredient at the rate of 1 pt. per 50 gals. (0.25% v/v). Use of crop oil is not advised since it may injure flowers and foliage. See Special Use Instructions. Sugar maples cannot be tapped for syrup within one year of application of this product. For repeat applications make on a minimum of a 14-day interval. Crop injury to non-bearing fruit and nut crops can occur if Clethodim 2E is improperly applied. Clethodim 2E must not be applied directly over the top of these plant types. Instead, spray must be directed at the base of the plant where grassy weeds are growing near the ground. RESTRICTIONS: - Aerial applications for all tree fruit uses are prohibited Clethodim 2E must not be applied to non-bearing fruit or nut crops which are grown for root stock. - If Clethodim 2E is applied as a spot treatment to non-bearing fruit and nut crops, DO NOT exceed the maximum rate allowed on a "per acre" basis. DO NOT apply more than 8 fl. oz./A (0.125 lb a.i./A) in a single application to non-bearing fruit and nut crops. DO NOT make more than 4 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.
Apples	<i>Malus</i> spp.		
Berries	<i>Vaccinium</i> spp.		
	<i>Rubus</i> spp.		
Cherry, Sweet	<i>Prunus avium</i>		
Citrus fruits	<i>Citrus</i> spp.		
Grapes	<i>Vitis</i> spp.		
Olives	<i>Olea</i> spp.		
Peach	<i>Prunus persica</i>		
Pears	<i>Pyrus communis</i>		
Prunes	<i>Prunus</i> spp.		
Stone Fruits	<i>Prunus</i> spp.		
Strawberries	<i>Fragaria</i> spp.		
Tree Nuts			
Almond	<i>Prunus triloba</i>		
Filbert	<i>Corylus illinoensis</i>		
Pistachio	<i>Pistacia vera</i>		
Walnut	<i>Juglans</i> spp.		

DIRECTIONS FOR USE IN ORNAMENTALS

Use Rate Per Acre	Special Use Instructions
6– 16 fl oz (0.094 – 0.25 lb ai)	For ornamental plant uses, Clethodim 2E can be used to control labeled grass weeds in greenhouses, lath houses, shadehouses, and around outdoor ornamentals, including nurseries, parks, roadside plantings, and structure landscapes. Add a non-ionic surfactant containing at least 80% active ingredient at the rate of 1 pt. per 50 gals. (0.25% v/v). Use of crop oil concentrate is not advised since it may injure flowers and foliage. For repeat applications make on a minimum of a 14-day interval. RESTRICTIONS: Clethodim 2E must not be applied to non-bearing fruit or nut crops which are grown for root stock. DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.

IMPORTANT

Clethodim 2E successfully controls weeds in newly transplanted and established non-grassy ornamentals. Plant resistance to **Clethodim 2E** at labeled rates has been found to be acceptable for the indicated genera and species listed below. Due to variability within species, crop growth stage, environmental conditions, and application techniques, it is advised that the user determine if herbicide can be used safely on a few plants prior to widespread application. Neither the seller nor the manufacturer of **Clethodim 2E** have investigated the safety factor to ornamental plants not listed on the label.

The following plants have shown a resistance for **Clethodim 2E** applications:

ORNAMENTAL TREES

COMMON NAME	SCIENTIFIC NAME
Alder, Red	<i>Alnus rubra</i>
Ash	<i>Fraxinus</i> spp.
Basswood	<i>Tilia</i> spp.
Birch, European White	<i>Betula pendula</i>
Birch, River	<i>Betula nigra</i>
Birch, White	<i>Betula papyrifera</i>
Crabapple, Flowering	<i>Malus halliana</i>
Dogwood, Flowering	<i>Comus florida</i>
Golden Chain Tree	<i>Laburnum anagyroides</i>
Maples	<i>Acer</i> spp.
Mulberry, White	<i>Morus alba</i>
Oaks	<i>Quercus</i> spp.
Olive, Wild	<i>Elaeagnus angustifolia</i>
Redbud, Eastern	<i>Cercis Canadensis</i>

Sweet Gum, American	<i>Liquidambar styraciflua</i>
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GROUND COVERS

COMMON NAME	SCIENTIFIC NAME
Bugleweed, Carpet	<i>Ajuga reptans</i>
Ivy, English	<i>Hedera helix</i>
Japanese Spurge	<i>Pachysandra terminalis</i>
Lilyturf	<i>Liriope muscari</i>
Moneywort	<i>Lysimachia nummularia</i>
Mondo Grass, White	<i>Ophiopogon jaburan</i>
Mondo Grass, Dwarf	<i>Ophiopogon japonicus</i>
Periwinkle, Lesser	<i>Vinca minor</i>

GARDEN FLOWERS AND PLANTS

COMMON NAME	SCIENTIFIC NAME
Ageratum	<i>Ageratum</i> spp.
Alyssum*, Sweet	<i>Lobularia maritima</i>
Asparagus Fern	<i>Asparagus setaceus</i>
Bleeding Heart	<i>Dicentra spectabilis</i>
Cast Iron Plant	<i>Aspidistra elatior</i>
Chrysanthemum	<i>Chrysanthemum</i> spp.
Cinquefoil	<i>Potentilla</i> spp.
Coleus	<i>Coleus</i> spp.
Coralbells	<i>Heuchera sanguinea</i>
Cranesbill	<i>Geranium</i> spp.
Dahlia	<i>Dahlia</i> spp.
Daisy, Trailing African	<i>Osteospermum fruticosum</i>
Daylily	<i>Hemerocallis</i> spp.
Dusty Miller	<i>Senecio cineraria</i>
Euonymus	<i>Euonymus</i> spp.
Gazania	<i>Gazania</i> spp.
Geranium, House	<i>Pelargonium hortorum</i>
Heather, False	<i>Cuphea hyssopifolia</i>
Hosta	<i>Hosta fortunei</i>
Iris	<i>Iris</i> spp.
Jasmine Tobacco	<i>Nicotiana glauca</i>
Loosestrife	<i>Lythrum salicaria</i>
Marigold	<i>Tagetes</i> spp.
Partridge berry	<i>Mitchella repens</i>
Petunia*	<i>Petunia hybrida</i>
Phlox	<i>Phlox</i> spp.
Pinks	<i>Dianthus</i> spp.
Portulaca	<i>Portulaca grandiflora</i>
Salvia	<i>Salvia</i> spp.
Saxifrage	<i>Saxifraga</i> spp.
Sedum	<i>Sedum</i> spp.
Selloum	<i>Philodendron selloum</i>

COMMON NAME	SCIENTIFIC NAME
Snapdragon*	<i>Antirrhinum majus</i>
Sweet Flag	<i>Acorus gramineus</i>
Tickseed	<i>Coreopsis grandiflora</i>
Touch-Me-Not	<i>Impatiens</i> spp.
Verbena	<i>Verbena</i> spp.
Violet	<i>Viola</i> spp.
Yarrow, Common	<i>Achillea millefolium</i>
Zinnia	<i>Zinnia elegans</i>

*Slight foliage or flower speckling has been observed on these species.

SHRUBS

COMMON NAME	SCIENTIFIC NAME
Abelia	<i>Abelia</i> spp.
Anise, Purple	<i>Illicium floridanum</i>
Aucuba	<i>Aucuba</i> spp.
Azalea*	<i>Rhododendron</i> spp.
Bamboo	<i>Bambusa</i> spp.
Barberry, Japanese	<i>Berberis thunbergii</i>
Barberry, Magellan	<i>Berberis buxifolia</i>
Bayberry	<i>Myrica pensylvanica</i>
Bottlebrush	<i>Callistemon citrinus</i>
Boxwood	<i>Buxus sempervirens</i>
Camellia	<i>Camellia japonica</i>
Candytuft	<i>Iberis sempervirens</i>
Cleyera	<i>Cleyera japonica</i>
Coralberry	<i>Ardisia crenata</i>
Crape Myrtle	<i>Lagerstroemia indica</i>
Coyote Brush	<i>Baccharis pilularis</i>
Fig, Creeping	<i>Ficus pumila</i>
Gardenia	<i>Gardenia</i> spp.
Holly	<i>Ilex</i> spp.
Honeysuckle	<i>Lonicera pileata</i>
Indian Hawthorn	<i>Raphiolepis indica</i>
Jasmine	<i>Jasminum</i> spp.
Jasmine, Asiatic	<i>Trachelospermum asiaticum</i>
Jasmine, Star	<i>Trachelospermum jasminoides</i>
Juniper	<i>Juniperus</i> spp.
Lantana	<i>Lantana</i> spp.
Nandina* Bamboo, Heavenly	<i>Nandinia domestica</i>
Oleander, Common	<i>Nerium oleander</i>
Oregon Grape	<i>Mahonia aquifolium</i>
Photinia	<i>Photinia</i> spp.
Pittosporum	<i>Pittosporum</i> spp.
Podocarpus	<i>Podocarpus</i> spp.
Privet	<i>Ligustrum</i> spp.
Pyracantha	<i>Pyracantha</i> spp.

COMMON NAME	SCIENTIFIC NAME
Rhododendron	<i>Rhododendron</i> spp.
Rose	<i>Rosa</i> spp.
Sweet Olive	<i>Osmanthus fragrans</i>
Viburnum	<i>Viburnum tinus</i>
Wisteria	<i>Wisteria</i> spp.
Yellow Sage/Shrub Verbena	<i>Lantana camara</i>

* Slight foliage or flower speckling has been observed on these species.

DIRECTIONS FOR ANNUAL GRASSES IN ORNAMENTALS

- Apply only to actively growing grasses at specified weed heights.
- Apply when the first grass weed species in a mixed grass weed population reaches the specified growth stage for treatment.
- Use the high rate under heavy grass pressure and/or when grasses are at maximum height.

GRASS SPECIES	SCIENTIFIC NAME	WEED* HEIGHT INCHES	RATE FL. OZ./ACRE¹ (LB A.I./ACRE)	HIGH RATE² (LB A.I./ACRE)
Barleygrass	<i>Echinochloa crus-galli</i>	2 to 8	8 [0.125]	16 [0.25]
Broadleaf Signalgrass	<i>Brachiaria platyphylla</i>	2 to 6	8 [0.125]	16 [0.25]
Brome				
California	<i>Bromus carinatus</i>	2 to 6	8 [0.125]	16 [0.25]
Cheatgrass	<i>Bromus secalinus</i>	2 to 6	8 [0.125]	16 [0.25]
Downy	<i>Bromus tectorum</i>	2 to 6	8 [0.125]	16 [0.25]
Ripgut	<i>Bromus diandrus</i>	2 to 6	8 [0.125]	16 [0.25]
Canarygrass	<i>Phalaris canariensis</i>	1 to 4	8 [0.125]	16 [0.25]
Crabgrass				
Hairy	<i>Digitaria adscendens</i>	2 to 6**	8 [0.125]	16 [0.25]
Large	<i>Digitaria sanguinalis</i>	2 to 6**	8 [0.125]	16 [0.25]
Smooth	<i>Digitaria ischaemum</i>	2 to 6**	8 [0.125]	16 [0.25]
Southern	<i>Digitaria ciliaris</i>	2 to 6**	8 [0.125]	16 [0.25]
Crowfootgrass	<i>Daetyloctenium aegyptium</i>	2 to 6**	8 [0.125]	16 [0.25]
Fall Panicum	<i>Panicum dichotomiflorum</i>	2 to 8	8 [0.125]	16 [0.25]
Field Sandbur	<i>Cenchrus incertus</i>	2 to 6	8 [0.125]	16 [0.25]
Foxtail				
Giant	<i>Setaria faberi</i>	2 to 12	8 [0.125]	16 [0.25]
Green	<i>Setaria viridis</i>	2 to 8	8 [0.125]	16 [0.25]
Yellow	<i>Setaria glauca</i>	2 to 8	8 [0.125]	16 [0.25]
Goosegrass	<i>Eleusine indica</i>	2 to 6**	8 [0.125]	16 [0.25]
Itchgrass	<i>Rottboellia exaltata</i>	2 to 6	8 [0.125]	16 [0.25]
Junglerice	<i>Echinochloa colona</i>	2 to 6	8 [0.125]	16 [0.25]
Lovegrass (Stinkgrass)	<i>Eragrostis cilianensis</i>	2 to 6	8 [0.125]	16 [0.25]
Rabbitsfootgrass	<i>Polypogon monspeliensis</i>	1 to 4	8 [0.125]	16 [0.25]
Red Rice	<i>Oryza sativa</i>	1 to 3	8 [0.125]	16 [0.25]
Ryegrass				
Hardy	<i>Lolium remotum</i>	2 to 6	8 [0.125]	16 [0.25]
Italian	<i>Lolium multiflorum</i>	2 to 6	8 [0.125]	16 [0.25]
Seedling Johnsongrass	<i>Sorghum halepense</i>	4 to 10	8 [0.125]	16 [0.25]

GRASS SPECIES	SCIENTIFIC NAME	WEED* HEIGHT INCHES	RATE FL. OZ./ACRE ¹ (LB A.I./ACRE)	HIGH RATE ² (LB A.I./ACRE)
Shattercane	<i>Sorghum bicolor</i>	6 to 18	8 [0.125]	16 [0.25]
Southwestern Cupgrass	<i>Eriochloa gracilis</i>	2 to 6	8 [0.125]	16 [0.25]
Sprangletop				
Amazon	<i>Leptochloa panicoides</i>	2 to 6	8 [0.125]	16 [0.25]
Bearded	<i>Leptochloa fascicularis</i>	2 to 6	8 [0.125]	16 [0.25]
Mexican	<i>Leptochloa uninervia</i>	2 to 6	8 [0.125]	16 [0.25]
Red	<i>Leptochloa filiformis</i>	2 to 6	8 [0.125]	16 [0.25]
Texas Panicum	<i>Panicum texanum</i>	2 to 6	8 [0.125]	16 [0.25]
Volunteer Cereals	<i>Hordeum vulgare</i>	2 to 6	8 [0.125]	16 [0.25]
Barley	<i>Hordeum vulgare</i>	2 to 6	8 [0.125]	16 [0.25]
Oats	<i>Avena sativa</i>	2 to 6	8 [0.125]	16 [0.25]
Rye	<i>Secale cereale</i>	2 to 6	8 [0.125]	16 [0.25]
Wheat	<i>Triticum aestivum</i>	2 to 6	8 [0.125]	16 [0.25]
Volunteer Corn	<i>Zea mays</i>	4 to 12	8 [0.125]	16 [0.25]
Volunteer Corn	<i>Zea mays</i>	12 to 24	8 [0.125]	16 [0.25]
Volunteer Grain Sorghum	<i>Sorghum bicolor</i>	8 to 12	8 [0.125]	16 [0.25]
Wild Oats	<i>Avena fatua</i>	2 to 6	8 [0.125]	16 [0.25]
Wild Proso Millet	<i>Panicum miliaceum</i>	2 to 10	8 [0.125]	16 [0.25]
Witchgrass	<i>Panicum capillare</i>	2 to 8	8 [0.125]	16 [0.25]
Woolly Cupgrass	<i>Eriochloa villosa</i>	2 to 8	8 [0.125]	16 [0.25]

*Generally occurs between 3-leaf stage and tillering.

**Length of lateral growth.

¹ 8 fl. oz./A=approximately 0.2 fl. oz./1000 sq. ft.

² 16 fl. oz./A=approximately 0.4 fl. oz./1000 sq. ft.

Add a non-ionic surfactant containing at least 80% active ingredient at the rate of 1 pt. per 50 gals. [0.25% v/v].

DIRECTIONS FOR ANNUAL BLUEGRASS CONTROL WITH CLETHODIM 2E IN ORNAMENTALS

GRASS SPECIES	WEED STAGE	RATE FL. OZ./A (LB A.I./ACRE)	HIGH RATE (LB A.I./ACRE)
Annual Bluegrass (<i>Poa annua</i>)	To 4-leaf	6 [0.09375]	16 [0.25]
Apply under favorable soil moisture and humidity which exists within a few days after rainfall or within 7 days after irrigation. Grass needs to be actively growing at time of application (s).			
Apply at weed stage indicated on the label, as reduced control can be expected with more mature annual bluegrass.			
Use high rate under heavy grass pressure and/or annual bluegrass is more mature.			
Add a non-ionic surfactant containing at least 80% active ingredient at the rate of 1 pt. per 50 gals. [0.25% v/v].			
For repeat applications make on a minimum of a 14-day interval.			
RESTRICTIONS:			
DO NOT apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application. DO NOT make more than 2 applications per acre per year. DO NOT apply more than 32 fl oz (0.500 lb ai) per acre per year.			

DIRECTIONS FOR PERENNIAL GRASSES

- Apply only to actively growing grasses at specified weed heights.
- Apply when first grass weed species in a mixed grass weed population reaches the specified growth stage for treatment.
- Use the high rate under heavy grass pressure and/or when grasses are at a maximum height.

RESTRICTIONS:

- **DO NOT** apply more than 16 fl. oz./A (0.25 lb a.i./A) in a single application.
- **DO NOT** make more than 2 applications per acre per year.
- **DO NOT** apply more than 32 fl oz (0.500 lb ai) per acre per year.
- For repeat applications make on a minimum of a 14-day interval.

GRASS SPECIES	WEED HEIGHT (Inches)	RATE FL. OZ./A ¹	HIGH RATE ²
Bermudagrass (<i>Cynodon dactylon</i>)			
First Application	3 (or up to 6" runners)	8 (0.125)	16 (0.25)
Repeat Application(s) (if re-growth occurs)	3 (or up to 6" runners)	8 (0.125)	16 (0.25)
Quackgrass (<i>Agropyron repens</i>)			
First Application	4 to 8	8 (0.125)	16 (0.25)
Repeat Application(s) (if re-growth occurs)	4 to 8	8 (0.125)	16 (0.25)
Rhizome Johsongrass (<i>Sorghum halepense</i>)			
First Application	12 to 24	8 (0.125)	16 (0.25)
Repeat Application(s) (if re-growth occurs)	6 to 18	8 (0.125)	16 (0.25)
Wirestem Muhly (<i>Muhlenbergia frondosa</i>)			
First Application	4 to 8	8 (0.125)	16 (0.25)
Repeat Application(s) (if re-growth occurs)	4 to 8	8 (0.125)	16 (0.25)

¹8 fl. oz./A=approximately 0.2 fl. oz./1000 sq. ft.

²16 fl. oz./A=approximately 0.4 fl. oz./1000 sq. ft.

Add non-ionic surfactant containing at least 80% active ingredient at the rate of 1 pt. per 50 gals. (0.25% v/v)

STORAGE AND DISPOSAL

Do not contaminate water, food, or feed by storage or disposal or cleaning of equipment. Open dumping is prohibited.

PESTICIDE STORAGE: Keep pesticide in original container. Do not put concentrate or dilute into food or drink containers. Store in cool, dry place. Do not store diluted spray.

PESTICIDE DISPOSAL: Wastes resulting from the use of this product may be disposed of on site or at an approved waste disposal facility.

CONTAINER HANDLING: Nonrefillable container. Do not reuse this container to hold materials other than pesticides or dilute pesticides (rinsate). After emptying and cleaning, it may be allowable to temporarily hold rinsate or other pesticide-related materials in the container. Contact your state regulatory agency to determine allowable practices in your state. Once cleaned, some agricultural plastic pesticide containers can be taken to a container collection site or picked up for recycling. To find the nearest site, contact your chemical dealer or manufacturer, or contact The Agricultural Container Recycling Council (ACRC) at www.acrecycle.org. If not recycled, then puncture and dispose of in a sanitary landfill, or incineration, or if allowed by state and local authorities, by burning. If burned, stay out of smoke. Triple rinse or pressure rinse container (or equivalent) promptly after emptying.

For packages up to 5 gallons: Triple rinse as follows: Empty the remaining contents into application equipment or a mix tank and drain for 10 seconds after the flow begins to drip. Fill the container $\frac{1}{4}$ full with water and recap. Shake for 10 seconds. Pour rinsate into application equipment or a mix tank or store rinsate for later use or disposal. Drain for 10 seconds after the flow begins to drip. Repeat this procedure two more times.

Pressure rinse as follows: Empty the remaining contents into application equipment or a mix tank and continue to drain for 10 seconds after the flow begins to drip. Hold container upside down over application equipment or mix tank or collect rinsate for later use or disposal. Insert pressure rinsing nozzle in the side of the container, and rinse at about 40 PSI for at least 30 seconds. Drain for 10 seconds after the flow begins to drip.

LIMITATION OF WARRANTY AND LIABILITY

NOTICE: Read the entire Directions for Use and Conditions of Sale and Limitation of Warranty and Liability before buying or using this product. If the terms are not acceptable, return the product at once, unopened, and the purchase price will be refunded.

The Directions for Use of this product must be followed carefully. It is impossible to eliminate all risks inherently associated with the use of this product. Crop injury, ineffectiveness or other unintended consequences may result because of such factors as manner of use or application, weather or crop conditions, presence of other materials or other influencing factors in the use of the product, which are beyond the control of TENKOZ, Inc. or Seller. To the extent permitted by applicable law, Buyer and User agree to hold TENKOZ and Seller harmless for any claims relating to such factors. TENKOZ warrants that this product conforms to the chemical description on the label and is reasonably fit for the purposes stated in the Directions for Use, subject to the inherent risks referred to above, when used in accordance with directions under normal use conditions. To the extent permitted by applicable law: (1) this warranty does not extend to the use of the product contrary to label instructions, or under conditions not reasonably foreseeable to or beyond the control of Seller or TENKOZ, and (2) Buyer and User assume the risk of any such use. TO THE EXTENT PERMITTED BY APPLICABLE LAW, TENKOZ MAKES NO WARRANTIES OF MERCHANTABILITY OR OF FITNESS FOR A PARTICULAR PURPOSE NOR ANY OTHER EXPRESS OR IMPLIED WARRANTY EXCEPT AS WARRANTED BY THIS LABEL. To the extent permitted by applicable law, in no event shall TENKOZ be liable for any incidental, consequential or special damages resulting from the use or handling of this product. TO THE EXTENT PERMITTED BY APPLICABLE LAW, THE EXCLUSIVE REMEDY OF THE USER OR BUYER, AND THE EXCLUSIVE LIABILITY OF TENKOZ AND SELLER FOR ANY AND ALL CLAIMS, LOSSES, INJURIES OR DAMAGES (INCLUDING CLAIMS BASED ON BREACH OF WARRANTY, CONTRACT, NEGLIGENCE, TORT, STRICT LIABILITY OR OTHERWISE) RESULTING FROM THE USE OR HANDLING OF THIS PRODUCT, SHALL BE THE RETURN OF THE PURCHASE PRICE OF THE PRODUCT OR, AT THE ELECTION OF TENKOZ OR SELLER, THE REPLACEMENT OF THE PRODUCT. TENKOZ and Seller offer this product, and Buyer and User accept it, subject to the foregoing Conditions of Sale and Limitation of Warranty and Liability, which may not be modified except by written agreement signed by a duly authorized representative of TENKOZ.

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